

Next-Track Session Recommendation: Which Trained Models Win, and Why

July 26, 2026

Abstract

This document compares the *trained* models built for the next-track task in this lab and argues why the successful ones succeed; the markdown reports in **experiments/** are the primary record and this synthesis transcribes rather than recomputes their numbers. The task is next-item *retrieval*: given a listening *session* (an ordered sequence of tracks), rank the 19 402-item catalog and recover the next *distinct* track the user actually played. The board is scored on one leak-free chronological split (**seq-20260715-131139**: 7154 sessions / 19 402 items, 5723 train / 1431 test) and is denominated in **Recall@10**, the record’s comparability currency; the server’s *declared* primary metric is **holisticness@10**, a four-factor list-quality composite that has had *three* incompatible code generations and that orders the models almost inversely, and this document keeps both boards distinct rather than silently choosing one. Six trained predictor families are in scope (Section 4); pure table lookups over the dataset appear only as reference bars and as blend *legs*, never as *compared* models. The one fact that explains most of the results is that 55 % of test targets are *cold* — never seen in any training session — which caps every seen-items-only mechanism at Recall@10 0.4493 and makes retrieval in a *learned content geometry*, not per-item memory, the only channel to the majority of the test set. On that reading the record has one champion — the promoted $R'+M+C'$ blend at Recall@10 0.2117 (303 of 1431), paired Δ over the unprojected three-leg crown +0.0259 [+0.012, +0.041], *CI entirely above zero*, seed-robust across three seeds and hardened on a second, colder split at +0.0426 [+0.0280, +0.0573] — a single-model bar of 0.12299 (176 of 1431) that no topology change has beaten *with a CI above zero*, five refuted design axes whose mechanisms are now *measured* rather than guessed (dual-tower view fusion, a ReLU pre-encoder, an MLP fusion head, four learned combiners, and depth/width/direction), and an eval-time re-ranking policy — maximal marginal relevance, MMR — that is the first honest lever on the crown and from which **nothing has been promoted**. The honest frame for the nulls is statistical: at $n = 1431$ the paired half-width is ~ 0.0124 on Recall@10 against a whole-family spread of 0.028, so large harms were reliably detected while *moderate wins were never detectable at all*.

1 Introduction: one split, one number, and why the cold half decides everything

Rationale. Every model in this document solves the same problem. A listening *session* is an ordered sequence of tracks; the model reads a prefix and must produce a ranking of the entire 19 402-item catalog such that the track the user actually played next lands near the top. Two separate conventions shape that, and the phrase “next-distinct” is easy to misread as covering both. First, the **target** is the next *distinct* track: an immediate repeat of something already in the prefix is skipped over when the label is constructed, so the task is not repeat detection in disguise — this is the operation the name refers to. Second, and independently, the **candidate**

set excludes every item already present in the prefix, which is the parity convention the Markov leg is built on; a model is therefore never credited for re-proposing a track the session already contains.

The one artifact — and the three places it is not. The board is scored on `seq-20260715-131139`, split *chronologically by session start* at 2024-08-24T14:28:02Z into 5723 train and 1431 test sessions, with every item represented by a frozen 192-dimensional latent (`spotify_tracks_song_pca192`). Three groups of numbers below are deliberately *not* on that artifact and are named at every use: the **seq-ann** pooled twin and one stacker arm (`seq-20260715-030509`, PCA-128), the second-split hardening of the champion (`seq-20260718-211238`), and the 2026-07-13/14 topology and first-blend campaigns, which predate it (`seq-20260713-014226`, built on the earlier *AE-64* item space — a 64-dimensional autoencoder embedding of the same track features, which a compression bake-off later replaced with PCA-192 because the linear projection retrieved better at every dimension tested). Their *verdicts* are paired Δ s against a same-split anchor and do travel; their *levels* do not, and must never be read beside the canonical-split column.

The hard fact. A test target is **cold** if its item index never appears in *any* of the 5723 training sessions. On this split 788 of 1431 targets are cold — a rate of 0.550664, which independently reproduces the manifest’s declared 0.5507. The consequence is a hard structural ceiling: any mechanism that can only retrieve items it observed during training — a co-occurrence table, a first-order Markov bigram, a free per-item embedding table — is capped at Recall@10 $643/1431 = 0.449336$ no matter how good it is. Everything above that line has to come from somewhere else, and on this corpus the only “somewhere else” is *content geometry*: a track that never appeared in training still has a feature vector, hence a position in the item space, hence a chance of being retrieved.

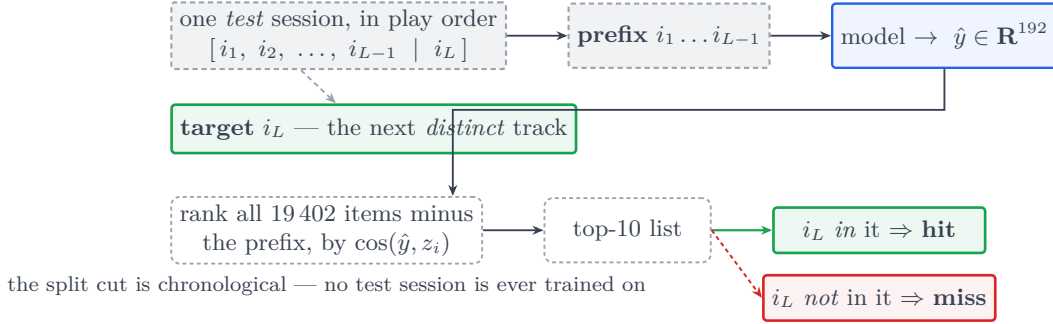
The second hard fact. The supervision budget is thin. Under per-step teacher forcing the 5723 training sessions yield 73 632 (prefix, next-item) pairs against a 19 402-item retrieval catalog, of which only 15 816 items are ever observed in training; the *median* observed item appears exactly *once*, the mean 5.017 times, and only 3294 items (17% of the catalog) reach five occurrences. These figures were recomputed here from the dataset artifact and agree with the DATA BUDGET block in the header of `predictors/seq_embed.py`. Read together with the cold rate they explain the shape of the entire record: every win on this board came from improving the *shared* latent geometry, and every attempt to spend parameters per item, per leg or per extra tower failed.

Interpretation. The rest of this document is one argument in that frame. Section 2 defines the machinery; Section 3 handles the awkward fact that this lab now has two primary metrics; Sections 4–5 lay out the trained models and their scores; Sections 6–9 explain the four things that work; Section 10 explains the five that do not, with a measured mechanism for each; Section 11 reframes those nulls as upper bounds; and Section 12 states what the record implies about where gains on this problem actually come from.

Sources: *PROJECT-FACTS.md*, and *2026-07-15c-nexttrack-phase2-model-sweep.md*.

2 Background: the machinery, defined once

This section is for a reader who knows mathematics and machine learning but not recommender-systems conventions. Nothing below is assumed later. Figure 1 draws the evaluation protocol every number in this document is produced by; the topology diagrams of Section 4 all presuppose it.



How every number in this document is produced. Repeat the above for each of the 1431 test sessions. *Recall@10* is the fraction that are hits (reported as “value (raw hits)”, e.g. 0.21174 (303)); *MRR* averages $1/\text{rank}(i_L)$ over the full ranking; the *graded* diagnostics ask instead whether some top-ten candidate shares i_L ’s artist or genre, and *music@10* how close the nearest of the ten is to i_L in the content metric M . Each session is additionally labelled **warm** or **cold** by whether i_L ’s item index occurs anywhere in the 5723 training sessions — 643 warm, 788 cold — which is the partition Section 6 turns on. During *training* the same picture is applied at *every* prefix position of every training session rather than only the last (“teacher forcing”, below), which is where the 73 632 supervision events come from.

Figure 1: The evaluation protocol, once. One test session is cut into an ordered prefix and a single held-out target; the model turns the prefix into one 192-dimensional point; the whole catalog minus the prefix is ranked by cosine to that point; the top ten either contains the target or does not. Everything else in this document is an aggregate over 1431 such trials, or a property of the ten returned items rather than of the target. Solid borders mark trained components, dashed borders frozen or parameter-free ones — the convention reused by Figures 4–9.

Cosine retrieval in a frozen item space. Every model here scores candidates the same way: it emits a single predicted vector $\hat{y} \in \mathbf{R}^{192}$ (or a full score column, for the lookup legs), and candidate i is ranked by $\cos(\hat{y}, z_i)$ against the *frozen* item latent z_i . The item table is not a parameter — it is read twice, once to embed the prefix and once as the retrieval target. This is why the geometry of that table is load-bearing, and why the single largest win on record is a learned *warp* of it (Section 7).

Teacher forcing (per-step supervision). For a session $[a, b, c, d]$ the supervised steps are $b \mid [a]$, $c \mid [a, b]$, $d \mid [a, b, c]$: at every timestep the model predicts the *next* item from the prefix so far, so one session of length L yields $L - 1$ training examples. On this split that is 73 632 supervision events rather than 5723. The distinction is not bookkeeping — two of the worst results on record are caused by an architecture that forced a switch to a one-target-per-session objective (Sections 10.4 and 10.5).

In-batch InfoNCE. The contrastive objective every trained model here uses. For a minibatch of B (prefix, target) pairs, embed the predictions a_i and the targets p_j , $L2$ -normalize both, form the $B \times B$ logit matrix $(a \cdot p^\top)/\tau$ and apply cross-entropy with the *diagonal* as the

labels. The $B - 1$ off-diagonal targets are the “in-batch negatives” — free negatives obtained by reusing every other row’s positive. The temperature τ (0.07 throughout unless stated) is the softmax sharpness: smaller τ demands a larger margin over the negatives. InfoNCE matters because ranking rewards a representation that *separates* the true next item from distractors, which a contrastive loss optimizes directly and a pointwise cosine regression does not.

The z-blend (score-level fusion). Each leg produces a full score column over the admissible candidates (all catalog items minus the prefix). Each column is z-scored *over that candidate set* and the columns are averaged with hard-coded weights: equal thirds for the three-leg recipe, $\alpha z(\text{GRU}) + (1 - \alpha)z(\text{Markov})$ with $\alpha = 0.5$ for the two-leg one. The z-scoring is the load-bearing detail: it puts legs whose raw scores live on incomparable scales onto a common per-session scale *while preserving relative magnitude*, so a leg that is confidently right about one candidate can outvote two diffusely uncertain legs.

Two spaces, and which one each quantity lives in. This is the single most confusing thing in the setup and it is stated once, here. **Retrieval** happens in the frozen PCA-192 item space: a candidate’s rank is $\cos(\hat{y}, z_i)$ with $z_i \in \mathbf{R}^{192}$. **Every list-quality facet**, by contrast, is a cosine in a *different*, larger space: the 517-dimensional intrinsic *content metric* M (the **balanced** dial of `spotify_tracks_content_metric`). `music@10`, `mood_coh@10`, `ild@10` and MMR’s diversity penalty are all cosines in M ; nothing in that group is computed in PCA-192. That split is exactly why the facets in Section 9 move together — one greedy penalty in M moves three quantities that are all defined in M — and why an eval-time re-rank can raise them while barely touching a ranking produced in PCA-192.

MMR (maximal marginal relevance) re-ranking. A purely *eval-time* policy: take the top `mmr_pool` candidates in base-score order, min-max normalize their scores to $\text{rel} \in [0, 1]$ over that pool, then build the top- k greedily by

$$\arg \max_i \lambda \text{rel}(i) - (1 - \lambda) \max_{j \in \text{selected}} \cos(M_i, M_j),$$

with M the 517-dimensional content metric just defined. In words: keep taking the most relevant remaining track, minus a penalty equal to its similarity to whichever already-chosen track it most resembles. The penalty is *continuous* — there is no similarity threshold and no cutoff radius; a highly relevant track can survive close to an already-picked one. $\lambda = 1$ recovers the base order exactly, so it is an exact identity gate; $\lambda \rightarrow 0$ ignores relevance and merely spreads the list out sonically. Crucially the *trained weights are bit-identical* across λ arms of one model, which is what makes those comparisons far sharper than architecture comparisons (Section 11).

Participation ratio. Used once, in Figure 19. Take the top candidates’ latents, weight each by its share of the score mass, centre them, and let $s_1 \geq s_2 \geq \dots$ be the resulting covariance eigenvalues normalized to sum to one. Then $\text{pr} = 1 / \sum_i s_i^2$ is the *effective number of directions* the belief spreads over: $\text{pr} = 1$ when all the variance is on one axis, and $\text{pr} = m$ when m directions carry equal shares. It is not entropy over items — a tight cluster of fifty near-identical tracks has high item entropy but $\text{pr} \approx 1$.

CKA (centered kernel alignment). A scalar in $[0, 1]$ measuring how similar two layers’ *representation geometries* are, computed from their centred Gram matrices; 1 means the two

carry the same pairwise-similarity structure up to rotation and scaling, 0 means unrelated. A middling value such as 0.458 therefore says the two representations are *partly* distinct, not that a fraction of them is new information.

Paired bootstrap. The decisive statistic throughout is a *paired* per-session Δ : resample the 1431 test sessions with replacement (2000 resamples, rng 1337), recompute both arms’ metrics on the *same* resampled index set, and report the 95% interval of the difference. A CI entirely above zero is a win, entirely below is a loss, and a CI straddling zero is a *tie*. Marginal (unpaired) CIs are also reported for the champion rows but are much wider and much weaker.

The ± 0.015 band — a policy constant, not a measurement. One number does two jobs in this record and it is worth separating them from the measured half-width. As a *practical tie band* it declares any Recall@10 difference under 0.015 not worth acting on; as an *anti-degenerate recall floor* it says a holisticness gain does not count unless Recall@10 stays within 0.015 of the reference. In both roles it is a chosen bright line, deliberately set *above* the measured cross-model paired half-width of ~ 0.0124 (Section 11) so that it is conservative rather than borderline. It is not pre-registered as a noise estimate and it should never be quoted as one — the measured per-comparison half-widths in this document range from 0.000382 to 0.018169 depending on the pairing structure.

Source: PROJECT-FACTS.md.

3 Two boards, three definitions: Recall@10 is the currency, holisticness@10 is the declared primary

Rationale. Two things are true at once and the document would be dishonest if it collapsed them. The experimental record’s *comparability currency* is Recall@10: the whole leaderboard is denominated in it, every paired decision rule is stated on it, and it is stable across every code change of the past month. The live server’s *declared* primary metric, however, is holisticness@10: `GET /api/best-models` returns `primary_metric = holisticness@10` and `domain.toml [metrics].primary` has read the same since 2026-07-23. The two orderings are close to inverted, so which board you read decides who is champion.

Recall@10 (↑). The fraction of test sessions whose held-out next track appears in the top ten retrieved. Reported throughout as “0.21174 (303)” — the value and the raw hit count out of 1431.

MRR (↑). Mean reciprocal rank of the true next track — the average of $1/\text{rank}$, so it rewards getting the truth to position 1 rather than position 9.

artist@10 / genre@10 (↑). *Graded* relevance — credit in degrees rather than the binary hit/miss of Recall@10. Both ask whether *some* top-ten candidate shares the true next track’s artist, respectively its primary genre. Exact recall is modest here by construction, and landing the right artist $\sim 33\%$ of the time and the right genre $\sim 40\%$ is what the models actually deliver. **These two columns must not be read as absolute quality.** Their chance floors are high and *have not been measured* on this corpus — there are far fewer distinct genres than items, so a random top ten hits the right genre vastly more often than it hits the right track. For contrast, chance Recall@10 is $10/19402 \approx 0.00052$, and the popularity lookup — which is not random but is the crudest real baseline — lands 1 hit in 1431 (Recall@10 0.00070) at genre@10 well above zero. Use artist@10 / genre@10 only to compare models against each other.

music@10 (↑). The *grounding* metric: the mean, over sessions, of the best cosine *in the* 517-dimensional content metric M (Section 2) between the *true* next track and the ten retrieved candidates. It equals 1.0 whenever the exact truth is in the list, so it is a smooth superset of Recall@10 that also credits landing something that merely *sounds like* the real continuation.

mood_coh@10 (↑). Mean cosine, again in M , of the top ten to the *prefix*’s mood centroid — the mean M -vector of the tracks already played. It asks whether the list belongs to the session it continues, and it can go *negative* for a list that is unrelated to the prefix.

ild@10 (↑). Intra-list diversity: mean pairwise $1 - \cos$ *within* the top ten, again in M . Note that this is exactly the quantity MMR’s penalty term (also a cosine in M) pushes down, which is why Section 9’s lever raises it by construction.

artist_adj@10 (↓). The “eagerness” or seed-echo rate: the fraction of the top ten sharing the artist of the *seed* track — the *last item of the prefix*, the track notionally still playing. It measures how much a ranker simply parrots what is on. **Lower is better.**

artist_conc@10 (↓). Normalized Herfindahl concentration over the top ten’s artists: with p_a the share of slots held by artist a and $n = 10$, $(\sum_a p_a^2 - 1/n)/(1 - 1/n)$, which is 0 when every slot is a different artist and 1 when one artist holds the whole list. It exists because artist_adj@10

is blind to a list of ten tracks by one *other* artist — a blind spot that is widest precisely on the pathological models. The out-of-scope **seq-mood** predictor (promoted under the definition name **mood-session**; called **seq-mood** throughout this document) reads `artist_adj` 0.003564, apparently “perfectly non-eager”, at `artist_conc` 0.329063 — a $\sim 90\times$ gap. Note that the *popularity* lookup also reads `artist_adj` ≈ 0.003 (0.002655), for the opposite reason: it returns the same ten globally popular tracks for every session and so echoes no seed at all. Two different pathologies, one apparently virtuous number. **Lower is better.**

The crown, and its three generations. `holisticness@10` multiplies four aggregate facets. The *live* (third) definition is

$$H = \text{clamp}(\text{mood_coh}, 0) \times \text{ild} \times (1 - \text{clamp}(\max(\text{artist_adj}, \text{artist_conc}), 0, 1)) \\ \times \text{clamp}\left(\frac{\text{music@10} - 0.34}{0.66}, 0, 1\right).$$

Three of those four factors are properties of the *returned list* rather than of its correctness, so a ranker that has given up on identifying the actual next track can maximize them all at once. That is not a fear, it is a measurement: under the original three-factor form $\text{clamp}(\text{mood_coh}, 0) \times \text{ild} \times (1 - \text{artist_adj})$ the out-of-scope **seq-mood** model held best-models rank 1 at H 0.2332 on Recall@10 0.0238 (34 of 1431), and a degenerate two-tower arm that reads only the slow session centroid took rank 2 at Recall@10 0.06639 (95 of 1431). Two fixes followed, both load-bearing: the *grounding* factor built on `music@10` (rescaled, because `music@10`’s raw spread across models is far narrower than the three-factor product’s, so multiplying in the raw value does not stop the ignore-the-user mode), and an anti-degenerate **recall floor** applied as a decision rule — an H gain does not count unless Recall@10 stays within ± 0.015 of the reference. Table 1 shows what those revisions did to identical predictions.

Results. On the live board the recall champion ranks *eighth of the nine* current best-models entries (H 0.008603, above only the two-leg R+M blend at 0.008259), while the leader is an eval-time re-ranking policy applied to the plain single GRU (H 0.034171 at 164 of 1431). Figure 2 plots the near-inversion; Figure 3 shows the failure mode the recall floor was written for.

Results — the two statements that look contradictory, reconciled. The record also says that the crown ordering carries *no information*, and both things are true because they are statements about different parts of the axis. Take the ten separately-trained architecture arms that carry a live H : they span 0.02127 to 0.02487, a spread of ~ 0.0036 against a measured $\text{hw}(\Delta H)$ of ~ 0.0018 — so within that *healthy band* the arms are statistically indistinguishable and their rank order is noise, exactly as PROJECT-FACTS.md states. The inversion is not inside that band. The recall champion sits at 0.008603, roughly seven half-widths *below* the band’s floor, and the MMR arms sit up at 0.026761 to 0.034171, well above its ceiling and resolved by a far sharper instrument (Section 11). So: the champion-versus-band gap and the band-versus-MMR gap are real and large; the ordering *within* the band is not a ranking of anything.

Interpretation. Picking one board fails in both directions: ranking by H alone crowns models with near-zero recall, while ranking by Recall@10 alone hides that the recall champion is the most artist-eager model measured (`artist_adj` 0.6097, `artist_conc` 0.6917 — inherited from its Markov leg, whose standalone `artist_adj` is 0.543, not from its neural one, since the single GRU and LSTM sit at 0.347 and 0.344). That the display facets are not merely “gameable by being good” is settled by the popularity floor: it *maximizes* `ild` (0.898) and *minimizes* `artist_adj` (0.002655, by echoing no seed at all rather than by being diverse) while landing one hit in 1431 and a *negative* mood

Table 1: The same predictions under all three `holisticness@10` generations — gen 1 (three-factor, to 2026-07-24), gen 2 (grounded four-factor with music floor 0.185, 2026-07-25), gen 3 (concentration-aware, eagerness becomes `max(artist_adj, artist_conc)` and the floor rises to 0.34; live since 2026-07-26). Values change by up to $6.4\times$ and *so does the ordering*: a stored `holisticness_at_k` is comparable only *within* one generation. This is survivable because every input facet is stored per row in each run’s `predictions.json`, so any generation is exactly recomputable for any past run — the generation labels used in this document were re-derived that way and matched by exact residual (0 to 3×10^{-7}). Recall@10 is given for scale. * run `run-20260726-161801-0df6f` and its bit-identical twin `run-20260725-143502-12fe6` are the same configuration with the same facets, and they *store* H 0.022841 and 0.146354 respectively — the cleanest single demonstration that a stored value carries its generation with it. Every cell here is a value that exists: gen-1 and gen-2 figures are as recorded in the campaign reports, gen-3 figures as returned by the live API. The only “—” in the table is the one genuinely uncomputable cell, the degenerate arm’s gen-3 value: that run is remote-only, never emitted `artist_conc_at_k`, and so cannot be rescored under a formula that needs it. Note also `seq-mood`’s gen-3 value: it is 0.0 *exactly*, because its music@10 of 0.33837 falls below the grounding floor 0.34 and the fourth factor clamps to zero. That single number is the grounding factor working as designed. *Sources: 2026-07-25-nexttrack-dual-tower-fusion-scan.md, 2026-07-26-nexttrack-pre-encoder-scan.md, and 2026-07-26-nexttrack-mmr-lambda-frontier.md.*

Run	R@10 (raw)	H gen 1	H gen 2	H gen 3 (live)
GRU h256 baseline	0.12299 (176)	0.141723	0.0470862	0.024538
GRU h425 (width control)	0.12089 (173)	0.13850	0.046677	0.024872
MLP256→GRU256 (S1)	0.10901 (156)	0.14499	0.047132	0.024102
Dual latent/latent, $f=1^*$	0.10273 (147)	0.146354	0.046612	0.022841
Dual cummean/cummean (degenerate)	0.06639 (95)	0.19715	0.050436	—
<code>seq-mood</code> (out of scope)	0.0238 (34)	0.23319	0.043881	0.0

coherence (−0.041). This document therefore denominates its board in Recall@10 and reports H beside it, always generation-tagged.

Sources: 2026-07-20-musical-distance-metric.md, 2026-07-22-session-holisticness-and-antieager-levers.md, and 2026-07-26-nexttrack-mmr-lambda-frontier.md.

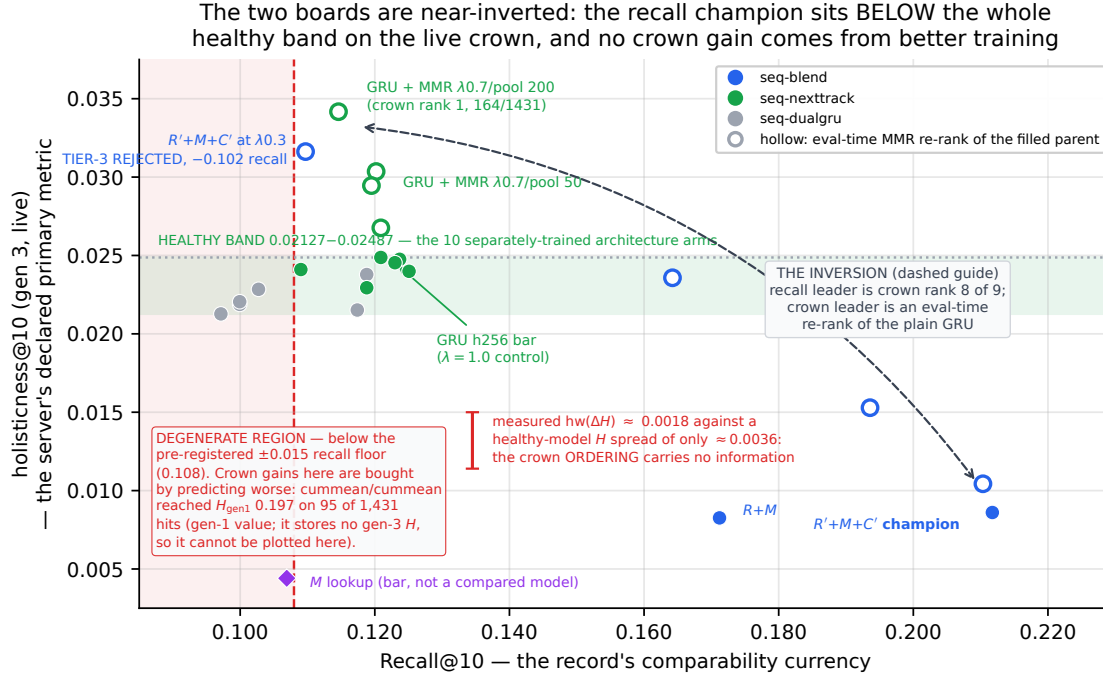


Figure 2: The two boards are nearly inverted *across* the healthy band, and carry no information *within* it. Recall@10 against *generation-3 (live)* holistictness@10 for the 22 distinct canonical-split configurations whose stored H is gen 3 (a further ten canonical-split runs are bit-identical repeats of these, and nine store gen-1 or gen-2 values and are excluded rather than rescaled). The recall champion sits bottom-right at H 0.008603; the crown leader is an eval-time MMR re-rank of the plain GRU at 0.034171 on 164 of 1431. MMR arms are drawn hollow because they share bit-identical trained weights with their parent — **none of them is promoted**, and the point labelled $\lambda 0.3$ on the champion trace is arm A4 of Table 3, a TIER-3 *rejected* arm that gets its high H by giving up 0.102 of recall; it is plotted to map the frontier, not as a candidate. The shaded green strip is the *healthy band* 0.02127 to 0.02487 spanned by the ten separately-trained architecture arms, and the drawn measurement bar is the point: that spread (~ 0.0036) is barely twice the measured $hw(\Delta H) \approx 0.0018$, so the ordering *inside* the strip is noise while the champion’s position far below it, and the MMR arms’ far above it, are not. The shaded red region below Recall@10 ≈ 0.108 is the degenerate zone the anti-degenerate recall floor excludes; the gen-1-only *cummean/cummean* arm lives there and is annotated rather than plotted, since rescaling it across generations would be fabrication.

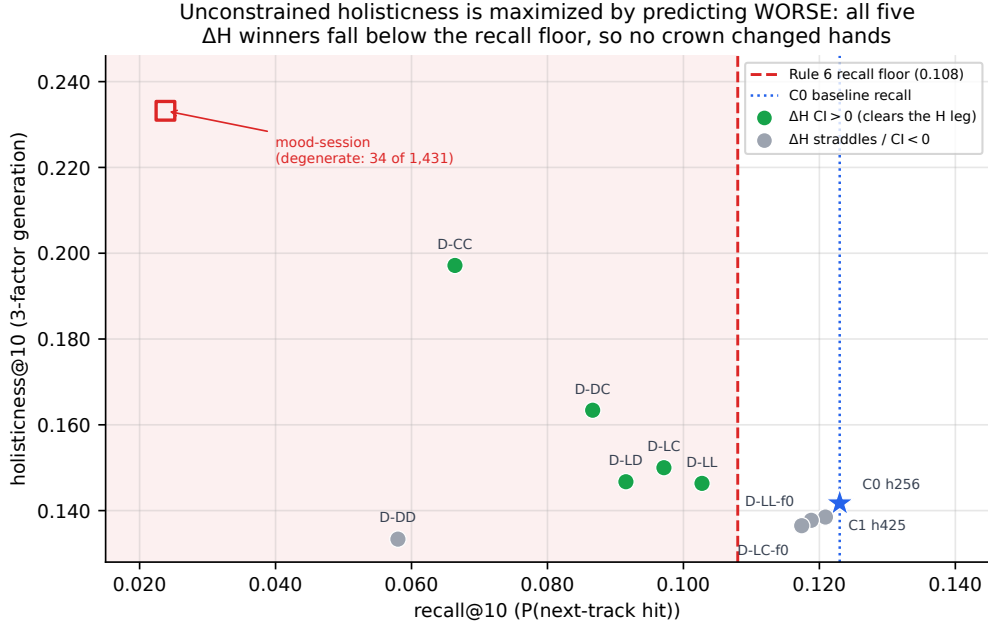


Figure 3: Why the crown needs a floor: unconstrained list-quality is maximized by predicting *worse*. **This figure is generation-1 (three-factor) holistiness** and its y values are not comparable to Figure 2’s or Table 1’s gen-3 column. Green circles are the five dual-tower arms whose paired ΔH CI clears zero; every one of them falls below the dashed anti-degenerate recall floor at $0.12299 - 0.015 = 0.10799$ (shaded), so no crown changed hands. The arrowed open square is the `seq-mood` incumbent at H 0.2332 on 34 hits of 1431 — the canonical instance of the pathology; the figure labels it by its promoted definition name, `mood-session`. Two labels in the figure are the source campaign’s internal shorthand: its legend’s “Rule 6” is that campaign’s numbering for the crown decision rule — an H gain counts only if Recall@10 holds within ± 0.015 — and the arm codes are C0 the single-GRU baseline, C1 the $h425$ width control, and D- XY the two-tower arm reading view X in tower A and view Y in tower B (L = `latent`, D = `delta`, C = `cummean`). *Source: 2026-07-25-nexttrack-dual-tower-fusion-scan.md.*

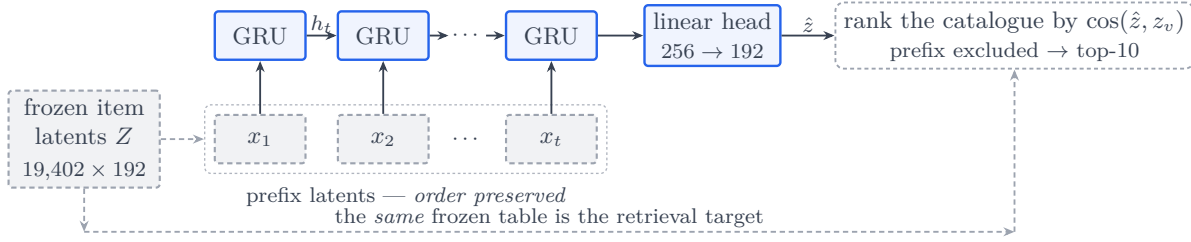
4 The model zoo: six topologies over one frozen geometry

Rationale. The six trained families differ in exactly one respect — *how the ordered prefix becomes one predicted point (or one score column)* — and are otherwise deliberately identical: same item space, same in-batch InfoNCE, same batching, same held-out-train-slice early stopping, same $\tau = 0.07$, same seed 1337. That is what makes the comparisons in Sections 6–10 clean. Figures 4–9 draw all six; the visual convention (solid border = trained, dashed = frozen or parameter-free, and colour carries the verdict) is fixed in Figure 4’s caption and used by the rest. All six presuppose the evaluation protocol of Figure 1: each one is a different answer to “how does the prefix become $\hat{y}$?”, and everything downstream of $\hat{y}$ is identical.

A one-line map before the diagrams. The six differ as follows. **seq-nexttrack** (Figure 4) runs a *recurrence* over the prefix latents and reads $\hat{y}$ off the final hidden state — the workhorse, and the single-model bar. **seq-ann** (Figure 5) replaces the recurrence with an order-blind *pooling* (mean of the prefix plus the last item) through an MLP — the control that isolates what recurrence buys. **seq-dualgru** (Figure 6) runs *two* independent recurrences over different causal views of the same prefix and fuses them — refuted, twice over, with the fusion head itself the larger harm. **seq-embed** (Figure 7) would make the item table *trainable* instead of frozen — the standard formulation in the literature, and the one family here with no runs at all. **seq-blend** (Figure 8) does not predict a point at all: it fuses three full-vocabulary *score columns* with fixed weights — the champion. **seq-stacker** (Figure 9) takes the same score columns and *learns* the combination with a gradient-boosted tree ensemble — the fourth of four learned-combiner nulls. Read the pair **seq-blend** versus **seq-stacker** as the document’s central contrast: same inputs, fixed versus learned combination, and a 0.13 recall gap in favour of fixed.

seq-embed: a design, not a result. One family in scope has no measurements at all and must be read as an argument rather than a row. `predictors/seq_embed.py` exists (written 2026-07-26) and implements a *learned* item-embedding table in three modes — **frozen**, **residual** and **free** — with InfoNCE forced because cosine has a trivial collapse optimum on a learned table. It is registered in `registry.toml` but does *not* appear in `GET /api/predictors`, has no model definition, and has **zero runs on any dataset**. Its interest is that the data budget of Section 1 predicts its outcome: a **free** table is the GRU4Rec/SASRec formulation, one trainable vector per item, estimated here from a *median of one observation*, and structurally capped at Recall@10 0.4493 because an unobserved item’s vector never receives a gradient. (That formulation — one free trainable vector per item, learned end-to-end from sequences — is the standard one in the session-recommendation literature, where GRU4Rec and SASRec are its recurrent and self-attentive canonical instances. It is the obvious thing to try and it is the one thing this corpus is least able to support.) This document quotes the dataset facts (73 632 steps, 15 816 observed items, median frequency 1, 3294 items at frequency ≥ 5) as measured, and everything about **seq-embed**’s performance as *unmeasured*.

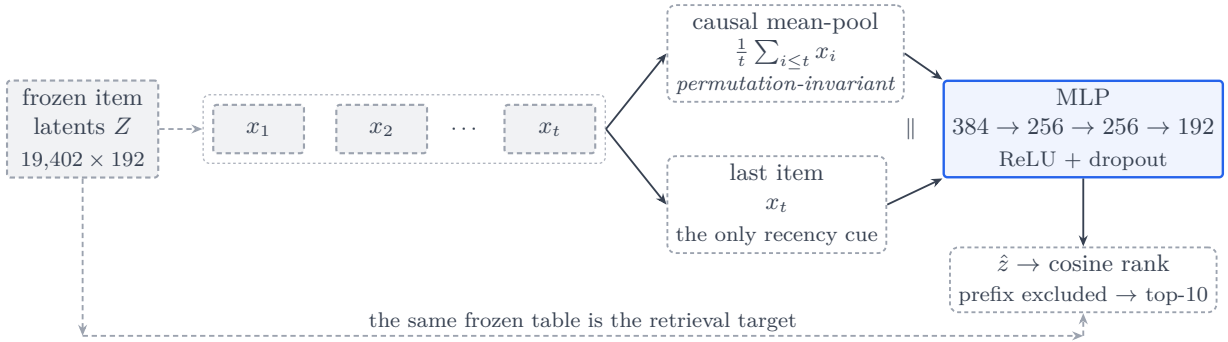
Sources: predictors/ sources read directly; PROJECT-FACTS.md.



Supervision — per-step teacher forcing. The head is shared across positions and must predict the next item’s latent at *every* prefix position, so one session of length L yields $L - 1$ examples: 73,632 (prefix, next-item) pairs on the train split. The loss is masked in-batch InfoNCE at $\tau = 0.07$ — each step’s true next latent must outscore the other steps’ targets in the same minibatch, which supplies negatives for free.

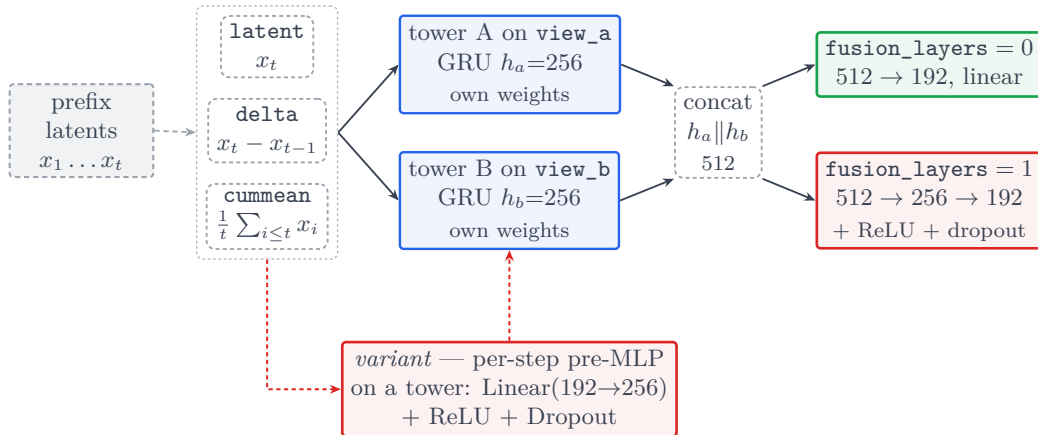
 trained (solid)
 frozen / parameter-free (dashed)
 table lookup
 parameter-free, learned space
 favoured variant
 refuted variant

Figure 4: **seq-nexttrack**, the single-model baseline every other topology in this document is measured against: a frozen item space, a recurrence over the *ordered* prefix, a linear head, and cosine retrieval back into the *same* frozen space. Nothing about the item geometry is learned here — the $19,402 \times 192$ latent table (offline PCA-192 on seq-20260715-131139) is read twice, once to embed the prefix and once as the retrieval target, and only the GRU and its linear head carry parameters (394 944 at $h = 256$). **The visual language used throughout this document is fixed here.** A *solid* border marks a trained component and a *dashed* border a frozen or parameter-free one; that distinction rides on the line style, not the colour, so it survives greyscale. Colour then carries the verdict — blue: the on-record default path; purple dashed: a non-parametric table lookup over the dataset; dashed blue: parameter-free but operating inside a *learned* geometry; green: a variant the record favours; red: a variant the record refutes. On the canonical split this configuration scores Recall@10 0.12299 (176 of 1431), reproduced bit-identically eleven times.



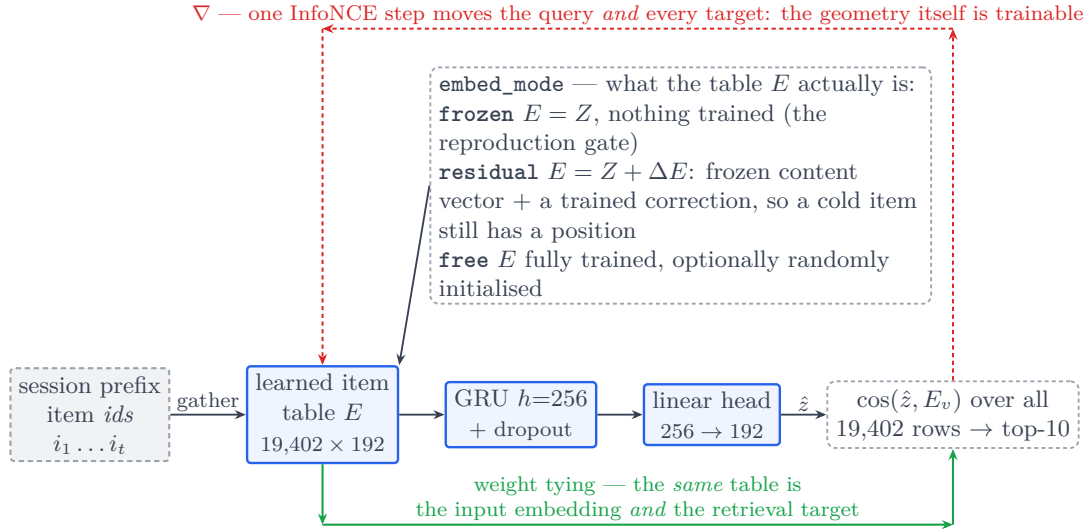
The same information, with the order discarded. Both summaries are strictly causal, so the objective, the padded batching and the early stopping are reused *verbatim* from Figure 4 — but a mean weights the track from twenty steps ago exactly like the one before last, and cannot represent “A then B” differently from “B then A”. Only the special-cased x_t tap carries any notion of recency, and the concatenation ($\parallel$) is this model’s entire account of sequence.

Figure 5: **seq-ann**, the GRU’s non-recurrent twin: the recurrence of Figure 4 is replaced by two parameter-free summaries of the same prefix — a causal mean-pool concatenated with the last item’s latent — feeding a plain three-layer ReLU MLP. Everything else is literally the same code (`seq_ann.py` imports `make_batches` and `step_loss` from `seq_nexttrack`), which is what makes this pair a clean test of the recurrence itself rather than of a training recipe. Visual convention as in Figure 4: the two pooling operators are dashed because they hold no parameters, and only the MLP is trained. The defensible comparison is same-space: on `seq-20260715-030509` (PCA-128) the pooled twin scores Recall@10 0.0901 (129 of 1431) against the GRU’s 0.1146 (164), paired $\Delta -0.0245$ $[-0.039, -0.011]$, a CI *entirely below zero*. **seq-ann** has no run at all on the canonical PCA-192 split, so it never belongs in the same column as the 0.12299 bar.



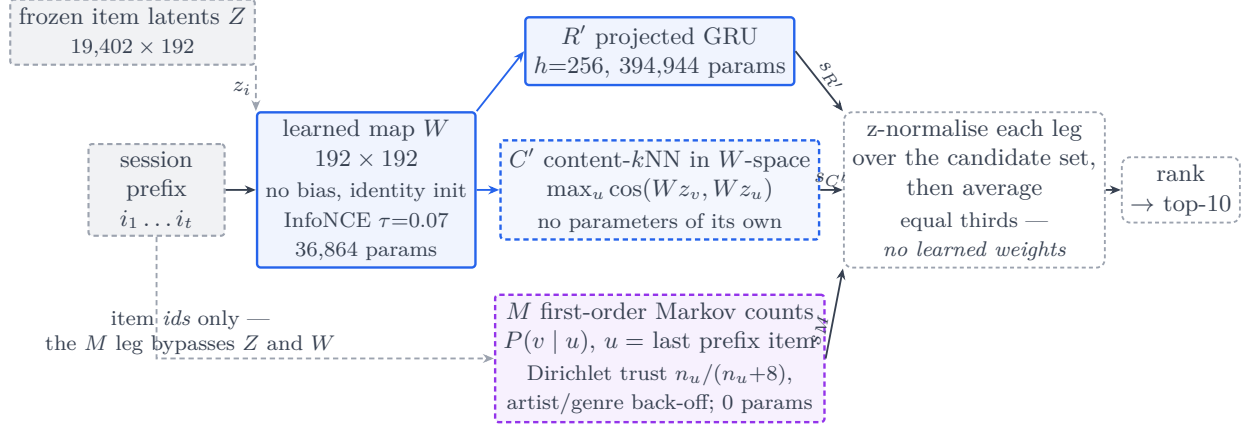
Both heads emit the same 192-d $\hat{z}$ and retrieve exactly as in Figure 4. The towers share nothing — separate modules, separate parameters — and their per-step hidden states are concatenated, so the combiner sees the towers’ *representations* rather than their scores and gradients flow back into both. There is no bidirectional mode by construction: a backward pass would let step t read item $t+1$ and copy its own teacher-forcing target. All three views are strictly causal for the same reason, and so is the pre-MLP, which is applied *timestep-wise*; it is drawn solid because it is trained, red because it is refuted.

Figure 6: **seq-dualgru**: two recurrent towers with entirely separate weights read the prefix in parallel and a head fuses their hidden states — the strongest available form of “train the combination end-to-end so the parts co-adapt”. Each tower’s input is a configurable causal view (**view_a** / **view_b**); the default is **latent/latent**, i.e. two independent GRUs over the same stream. Two of the boxes are coloured by the record rather than by the design. Removing the MLP fusion head (**fusion_layers** = 0, green) *beats* keeping it in three independent measurements, most recently by +0.02655 [+0.01398, +0.03913] CI>0 — and with some 9–10% *fewer* parameters, so the linear readout is the family default; and the per-step pre-MLP (red) loses the decisive single-tower ablation, Recall@10 0.10901 (156 of 1431) against the baseline’s 176, paired Δ −0.01398 [−0.02657, −0.00140] CI<0. Visual convention as in Figure 4. The best dual arm reaches 0.11880 and merely *ties* the single-GRU bar, as does a 0.098% parameter-matched width control — so capacity is not the explanation.



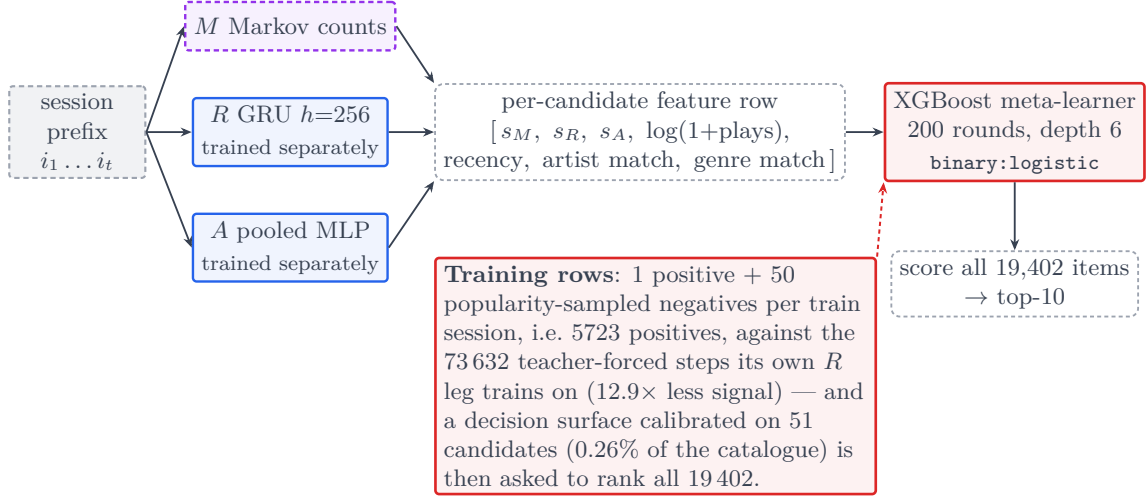
Why the loop is the whole point. In Figures 4–6 the retrieval space is fixed and only the map into it is learned; here the space is a parameter, shaped by next-track adjacency rather than inherited from a reconstruction objective. That also creates a collapse hazard absent everywhere else: with a learned table a plain cosine objective has a trivial optimum at “send every item to one point”, so only a contrastive loss is admissible and `loss=cosine` is rejected outright for the trained modes.

Figure 7: **seq-embed**, the one axis the record has not tested: **a design, not a result**. No run of this predictor exists on any dataset, it is absent from the server’s registered predictor list, and nothing here is a measured score. The item table E is used twice, as the GRU’s input embedding and as the retrieval target (weight tying, the GRU4Rec/SASRec formulation), so a single gradient step moves the query and all the candidates together — the return arrow along the top is what makes the geometry trainable, and it is exactly what every other topology in this document lacks. The three **embed_mode** settings span the space from a reproduction gate to a fully free table. What *is* measured is the data budget that bounds them: 73 632 training steps to fit 15 816 item vectors, a median per-item training frequency of 1, only 3294 items occurring five or more times, and 788 of 1431 test targets never appearing in any training session — so a **free** table cannot retrieve the cold 55 % at all and is capped at Recall@10 0.4493, whereas **residual** keeps the content vector and with it the cold reach. Visual convention as in Figure 4.



Which legs are learned. W and the R' recurrence are trained (431 808 parameters between them); C' has no parameters of its own but retrieves inside W 's learned geometry, which is why it is drawn dashed-blue; M is a table of training-set transition counts, drawn dashed-purple. The combiner is a fixed rule: each leg's full-vocabulary score vector is z-scored over the admissible candidates (every item minus those already in the prefix) and the three are averaged with hard-coded equal weights. The z-scoring is the load-bearing step — it puts three incomparable score scales onto one per-session scale while *preserving* relative magnitude, so a confidently-peaked leg can outvote two diffuse ones. Rank-only fusion, which discards that magnitude, loses by -0.029 CI <0 .

Figure 8: **seq-blend** with **projection=true**, the champion: three full-vocabulary score vectors, z-normalised over the candidate set and averaged in fixed equal thirds. W acts on the *frozen item latents* Z of Figure 4 — it warps the retrieval geometry, it does not read the prefix ids directly — and it is shared by two of the three legs and bypassed by the third; that is the whole mechanism. W is fit by in-batch InfoNCE on consecutive within-train pairs, which rebuilds the retrieval metric from “reconstruct the item matrix” into “what follows what”; both legs that retrieve by cosine in that space improve — the parameter-free content- k NN leg more than doubles, 0.0685 to 0.1516, and the recurrent leg rises from 0.1230 to ≈ 0.172 — while the item-ID lookup is untouched at 0.106918. The two projected-leg standalone figures come from the off-server driver of the 2026-07-18 campaign and have no server run of their own; only the M leg's number is a canonical-split run. Together the three legs reach Recall@10 0.21174 (303 of 1431), paired Δ over the unprojected three-leg blend $+0.0259$ [$+0.012$, $+0.041$], a CI entirely above zero and robust across three seeds. Visual convention as in Figure 4.



Why the shape matters. A 200-round depth-6 booster is piecewise *constant*, so ranking 19 402 candidates with it produces large tied blocks that the harness breaks by item index. The fixed z-blend of Figure 8 is instead a strictly monotone continuous function of the leg scores: it preserves each leg’s internal ordering exactly and can only trade the legs off against one another.

Figure 9: **seq-stacker**, the learned counterpart of Figure 8’s fixed rule: the same separately-trained base legs, but their per-candidate scores become columns of a feature row alongside four cheap candidate features, and a gradient-boosted tree learns the combination. It earns a diagram because its failure is informative — with the **markov**, **gru** and **ann** legs it reaches Recall@10 0.085255 (122 of 1431), *below its own GRU base leg* at 0.12299, paired $\Delta -0.0377$ $[-0.056, -0.020]$ CI<0. The red callout names the two mechanisms: supervision starvation, since the meta-learner gets one labelled event per session where its base leg gets one per prefix position, and a train/serve candidate mismatch. The booster is drawn solid because it is trained and red because it is refuted; the feature row and the ranking step are dashed because they hold no parameters. Visual convention as in Figure 4. This is one of four independent learned-combiner nulls, the others being reciprocal-rank fusion, oracle weights fitted on the test labels, and the jointly-trained dual tower of Figure 6.

5 The board: what the trained models actually score

Results. Table 2 is the in-scope board and Figure 10 is the same information as a landscape, with the two non-parametric reference bars drawn as *lines* rather than compared rows. Figure 11 shows the family ordering that first established it. Three reading rules apply.

First, the promoted champion is the $R'+M+C'$ blend at 0.21174 (303 of 1431); the 0.22292 (319) cell above it is a temperature scan point ($\tau = 0.10$) whose +0.011 nominal edge was judged *sub-noise* and whose default was **held** — it is the highest Recall@10 ever measured on this split and it is *not* the champion’s number.

Second, the single-model bar is the plain GRU at 0.12299, which has been reproduced *eleven* times bit-identically ($0.12299091544374564 = 176/1431$, MRR agreeing to 3.1×10^{-11}) across hosts and across a predictor code edit. Same-configuration reproduction spread on this deterministic path is therefore exactly zero, which is why the noise discussion in Section 11 is about *between*-configuration jitter, not seeds.

Third, **seq-ann**’s only run in the entire system is on **seq-20260715-030509** (PCA-128), *not* the canonical split. Because both splits share the same session index its n_{test} is also 1431, which makes the number look canonical when it is not; it is therefore daggered and excluded from the best-cell bolding, and the defensible GRU-versus-ANN comparison is the same-space one in Section 8. Seven further cells carry a §: they are *nominal single-run* comparisons for which no paired bootstrap CI was ever computed — the two temperature cells, the projection-rank cell, the two BPR cells, the anti-eager β arm and the canonical-split LSTM. The reports call some of those ties and one a collapse; none of those calls is backed by a paired test, and none of the cells is eligible for a best-cell bold.

Table 2: The in-scope trained models on the canonical split `seq-20260715-131139` (1431 test sessions), sorted by Recall@10. Eval-time MMR arms are held back to Table 3 because they share bit-identical weights with their parent. The *family* column abbreviates the predictor: `blend` = `seq-blend`, `next` = `seq-nexttrack`, `dual` = `seq-dualgru`, `ann` = `seq-ann`, `stack` = `seq-stacker`, `embed` = `seq-embed`, `lookup` = a non-parametric table over the dataset. **Directions:** R@10, MRR, art@10, gen@10 and *H* are all higher-better. *H* is the *generation-3 (live)* composite and is **not** comparable to Table 1’s gen-1/gen-2 columns (Section 3); the champion’s low *H* (0.008603, the smallest non-lookup value in the column) is the two-board inversion of Section 3, not a defect, and it is not resolvable against the 0.02127 to 0.02487 healthy band by anything in this table. For absolute scale, chance Recall@10 for a random top ten is 10/19402 $\approx$ 0.00052 and the popularity lookup reaches 0.00070 (1 hit); the art@10 and gen@10 chance floors are much higher and have *not* been measured, so read those two columns only across rows. **Bolding.** Two rows are bolded, the promoted champion and the prior crown it replaced. Best cell per metric is bolded among the *on-split*, *unrefuted*, *non-nominal* rows only — which excludes the `seq-ann` row (off-split), the Markov row (a lookup, kept in sorted position for scale rather than as a compared model), the τ 0.10 scan cell and the other $\S$ -marked cells (nominal, no paired CI), and every `seq-dualgru`, pre-encoder, stacker and β arm shown to *bound* a refuted axis. On that rule the champion holds R@10 and MRR and the *h425* width control holds art@10, gen@10 and *H*. “—” marks a value not recorded: the nine remote-only runs never emitted `artist_conc_at_k` and so could not be rescored to generation 3 (their stored *H* is gen 1 or gen 2 and is therefore withheld rather than printed in a gen-3 column), and no parameter count is recorded anywhere for `seq-stacker` or `seq-ann`. $\dagger$ *not* on the canonical split — `seq-20260715-030509`, PCA-128; do not read it in this column. $\ddagger$ scan cell whose nominal edge was judged sub-noise; the $\tau = 0.07$ default was held. $\S$ *nominal single-run comparison* — no paired bootstrap CI exists for this cell (Section 13). *Sources:* `2026-07-15c-nexttrack-phase2-model-sweep.md`, `2026-07-18b-nexttrack-projection-registration-and-scan.md`, `2026-07-25-nexttrack-dual-tower-fusion-scan.md`, `2026-07-26-nexttrack-pre-encoder-scan.md`, and `2026-07-26-nexttrack-mmr-lambda-frontier.md`.

Model / arm	family	params	R@10 (raw)	MRR	art@10	gen@10	<i>H</i> gen 3	Note
proj-blend τ 0.10 $\ddagger\S$	blend	431808	0.22292 (319)	0.1273	0.3424	0.4032	—	held sub-noise
<i>R'</i> + <i>M</i> + <i>C'</i> τ 0.07	blend	431808	0.21174 (303)	0.1220	0.3326	0.4011	0.008603	promoted, servable
proj-blend rank-64 $\S$	blend	407232	0.20545 (294)	0.1169	0.3326	0.3990	—	rank is not the gain
proj-blend τ 0.05 $\S$	blend	431808	0.19287 (276)	0.1111	0.3354	0.3997	—	τ ablation
R + M + C	blend	394944	0.18588 (266)	0.1025	0.3305	0.3941	—	prior crown
R+M two-leg	blend	394944	0.17121 (245)	0.0952	0.3291	0.3983	0.008259	leg-ablation ref.
proj-blend BPR full $\S$	blend	431808	0.15793 (226)	0.0939	0.3187	0.3913	—	InfoNCE $\gg$ BPR
proj-blend BPR rk-64 $\S$	blend	407232	0.15234 (218)	0.0909	0.3194	0.3906	—	worst proj. cell
C2 GRU h454 <i>ctrl</i>	next	969936	0.12509 (179)	0.0539	0.3375	0.4542	0.023988	width ctrl, tie
GRU h256 $\beta=0.05\S$	next	394944	0.12369 (177)	0.0534	0.3368	0.4500	0.024755	weak knob
T1-f0 bare/preMLP256	dual	888256	0.12369 (177)	0.0547	0.3333	0.4647	—	a repair, not a win
GRU h256 (bar)	next	394944	0.12299 (176)	0.0534	0.3375	0.4500	0.024538	single-model bar
C1 GRU h425 <i>ctrl</i>	next	871017	0.12089 (173)	0.0522	0.3410	0.4577	0.024872	tie, 2.2 $\times$ params
LSTM h256 $\S$	next	510144	0.11880 (170)	0.0493	0.3277	0.4549	0.022941	GRU $\approx$ LSTM
D-LL-f0 latent/latent	dual	789696	0.11880 (170)	0.0563	0.3368	0.4563	0.023785	best 07-25 dual: tie
GRU h297 <i>ctrl</i>	next	494697	0.11321 (162)	0.0520	0.3284	0.4500	—	outlier, CI<0
S1 MLP256 $\rightarrow$ GRU256	next	493504	0.10901 (156)	0.0446	0.3235	0.4689	0.024102	pre-encoder harmful
B2 first-order Markov	lookup	0	0.10692 (153)	0.0694	0.2984	0.3885	0.004409	reference bar only
D-LL latent/latent, <i>f</i> 1	dual	871872	0.10273 (147)	0.0424	0.3089	0.4577	0.022841	fusion head harmful
T4 preMLP both towers	dual	1068992	0.09783 (140)	0.0383	0.2991	0.4389	—	placement irrelevant
pooled-MLP twin $\dagger$	ann	—	0.09015 (129)	0.0394	0.2991	0.4354	—	off-split, PCA-128
D-DC delta/cummean	dual	871872	0.08665 (124)	0.0338	0.2579	0.4242	—	beats both parents
XGB stacker (M+G+A)	stack	—	0.08526 (122)	0.0490	0.3005	0.3753	—	below its base leg
D-CC cummean/cummean	dual	871872	0.06639 (95)	0.0266	0.2460	0.4060	—	degenerate arm
D-DD delta/delta	dual	871872	0.05800 (83)	0.0282	0.2523	0.4228	—	worst real arm
learned item table	embed	—	—	—	—	—	—	design only, no runs

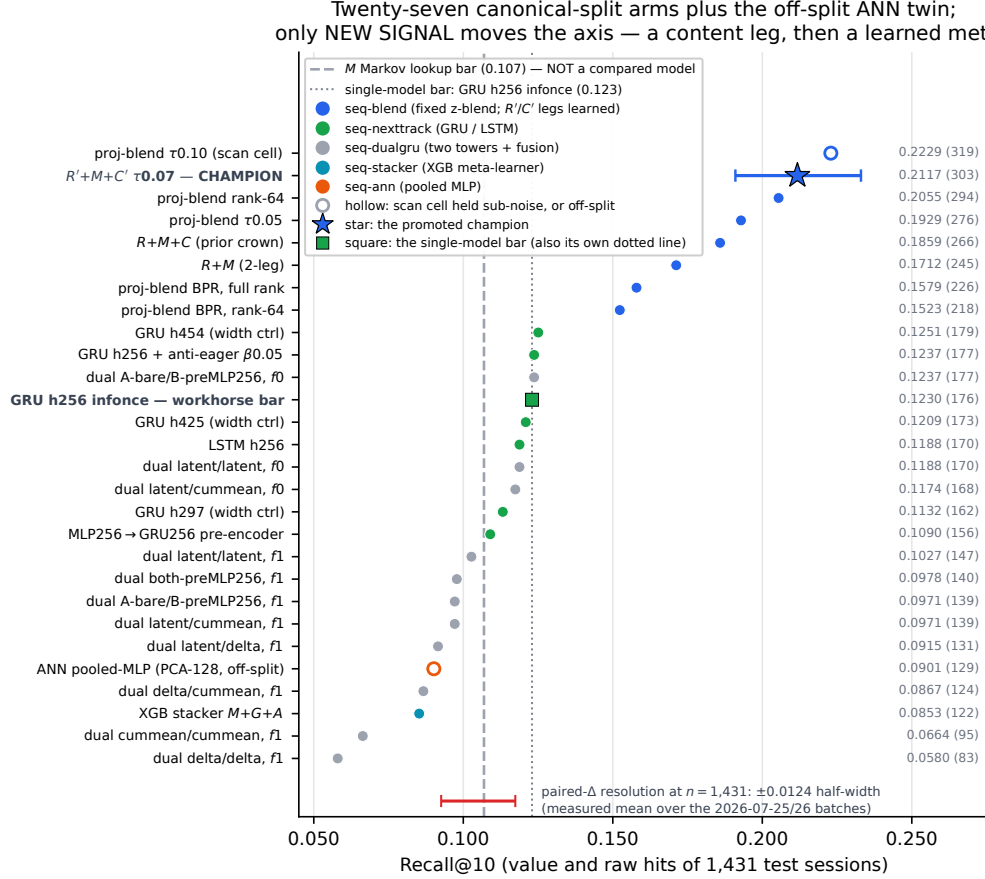


Figure 10: The whole in-scope landscape, and the things it is easy to misread. 27 trained arms on the canonical split plus the off-split ANN twin (28 rows in all), ordered by Recall@10 with raw hits of 1431 and coloured by predictor family (**seq-blend** blue, **seq-nexttrack** green, **seq-dualgru** gray, **seq-stacker** cyan, **seq-ann** orange). The champion is the blue star. The first-order Markov lookup (0.107) is drawn only as a reference *line*, never as a row, because it is a table over the dataset; the single-model GRU bar (0.123) is drawn *both* as its own dotted line and as a row (bold label, green square), because it is simultaneously a compared model and the anchor every architecture Δ is measured against. Two markers are hollow for integrity: the ANN row because it is PCA-128 off-split, and the $\tau 0.10$ cell because its nominal edge was held sub-noise — it is the highest Recall@10 on the split and it is *not* the champion. Only the champion carries a recorded marginal 95% CI [0.191, 0.233]; the separate ± 0.0124 bar states the paired- Δ resolution of the instrument, which is what actually decides the adjacent rows.

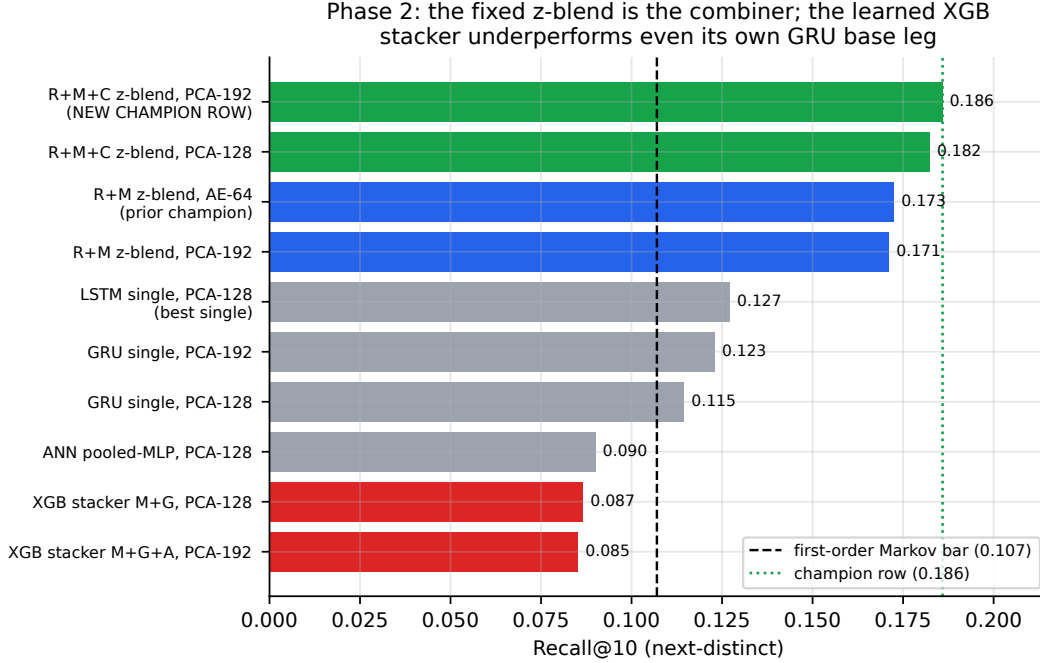


Figure 11: The family ordering that set the shape of the board, from the Phase-2 sweep. Green: the *fixed* z-blends. **Read the in-figure label “NEW CHAMPION ROW” historically:** the three-leg R+M+C on PCA-192 at 0.1859 was the champion at the date of that sweep and is the *prior* crown here, superseded by $R'+M+C'$ at 0.21174 (Section 7), which is not in this figure. Blue: other fixed blends. Gray: single models — the LSTM on PCA-128 (0.1272) is the best single model in that sweep, the pooled ANN twin (0.0901) the worst. Red: the *learned* XGBoost stackers at 0.0853 to 0.0867, below every single model they were built on top of. The first-order Markov bar (0.107) is the incumbent reference line. Read left to right the figure already contains the document’s thesis: fixed fusion of unlike signals beats every single model, and learning the fusion loses. *Source: 2026-07-15c-nexttrack-phase2-model-sweep.md.*

6 Fusion wins because its legs are correct on almost disjoint sessions

Rationale. The champion is a score-level fusion of three legs, and the naive story — “ensembling reduces variance” — is wrong here in an instructive way. What each leg *knows* is structurally different. The Markov leg is a pure count lookup keyed on the single last prefix item u : the empirical $P(v \mid u)$ over training bigrams. Its counts are shrunk toward a fallback by a *trust weight* $n_u/(n_u+8)$, where n_u is how many times u was ever followed by anything in training — so an item seen once is trusted 1/9 and an item seen a hundred times almost fully, and the remaining mass goes to an artist/genre back-off distribution plus a tiny popularity tie-break. (This is Dirichlet-multinomial smoothing with a pseudo-count of 8; the point is only that a leg built on counts must have *something* to say when the counts are thin.) It can only score items it has *seen follow something*. The GRU leg regresses a point in the item space from the ordered prefix and ranks the whole catalog by cosine to it; it never needs to have seen the target at all, only to land near it. The content-kNN leg is the same idea with no training (max cosine of a candidate to any prefix item). So the fusion is not two estimates of one quantity, it is a *union of two retrieval mechanisms* — identity memory and content geometry — whose failure modes are close to complementary.

Results. The warm/cold decomposition makes the mechanism visible. Partitioning the 1431 test targets into 643 warm and 788 cold by the train-seen test and recomputing recall in each half from each run’s `predictions.json` gives: popularity 0.0016 warm / 0.0000 cold; Markov 0.2255 (145/643) warm / 0.0102 (8/788) cold; the GRU 0.1198 (77/643) warm / 0.1256 (99/788) cold; R+M 0.2255 / 0.1269; R+M+C 0.2255 / 0.1536; and the champion 0.2395 (154/643) / 0.1891 (149/788). Figure 12 plots this. The asymmetry is extreme: Markov is 22× better on warm targets than cold ones, while the GRU is *flat* across the split and in fact very slightly better on cold — exactly what a content-space retrieval should be. Hit-set overlap on the identical sessions: $|M| = 153$, $|R| = 176$, $|M \cap R| = 55$, $|M \cup R| = 274$, Jaccard 0.201; 98 sessions are Markov-only and 121 GRU-only. R+M lands 245 hits, 211 of them inside that 274-session union plus 34 sessions *neither* leg hit alone. Note the scope carefully: this overlap argument is about the **R+M stage**, computed with the *unprojected* GRU leg R . The champion’s 303 hits do include 109 sessions outside $M \cup R$ entirely, but that number mixes the fusion effect with the projection upgrade of Section 7 — it is not evidence about fusion, and the champion’s extra reach is measured there, not here. The fusion win itself is CI-backed. On the AE-64-lineage artifact `seq-20260713-014226` where the two-leg blend was first established, paired Δ over the Markov bar on that split is +0.0657 [+0.049, +0.083], three-seed confirmed; on the canonical split the content third leg is +0.0147 [+0.003, +0.026] over same-space R+M, both CIs above zero. The warm/cold and overlap counts are *single-run observations with no CI*, though they reproduce independently-reported off-server figures and rest on deterministic runs.

Interpretation — and three alternatives ruled out. (a) “More models help.” No: adding a leg of the *same kind* does nothing. *All four numbers in this paragraph are on the 2026-07-13/14 AE-64-lineage artifact seq-20260713-014226, not the canonical split, and must be read only against each other* — the single GRU on that artifact scores ≈ 0.089 , not 0.12299. There, adding the pooled ANN as a third leg gives R+A+M 0.1684 against R+M’s 0.1726, i.e. slightly worse, and R+A alone is 0.0894, essentially that split’s GRU alone. Substituting *item2vec* — a skip-gram co-listening embedding, the word2vec construction applied to sessions-as-sentences — for the content leg gives R+M+I 0.1349, well below R+M. And on the canonical split a fourth leg of *session-kNN* type (retrieve whole *sessions* similar to the current one and vote over what followed them; V-SKNN in the literature) drags R+M+C+V to 0.1474, paired Δ −0.038 CI below zero — and it fails for

precisely the mechanism above, being structurally warm-only against a 55% cold test set in a single-user corpus where there is no cross-user neighbourhood to borrow from. Diversity of signal *kind*, not model count, is what pays. (b) “The blend just inherits Markov’s warm hits.” No: R+M lands 100 cold hits where Markov alone lands 8, and the entire cold half of the champion’s performance (149 of 303) is carried by legs Markov contributes nothing to. (c) “A smarter fuser would do better.” That is Section 10.4, and it is five independent nulls.

Sources: *2026-07-14-nexttrack-blend-and-ensemble.md*, *2026-07-15c-nexttrack-phase2-model-sweep.md*, and *2026-07-18-nexttrack-literature-fit-campaign.md*.

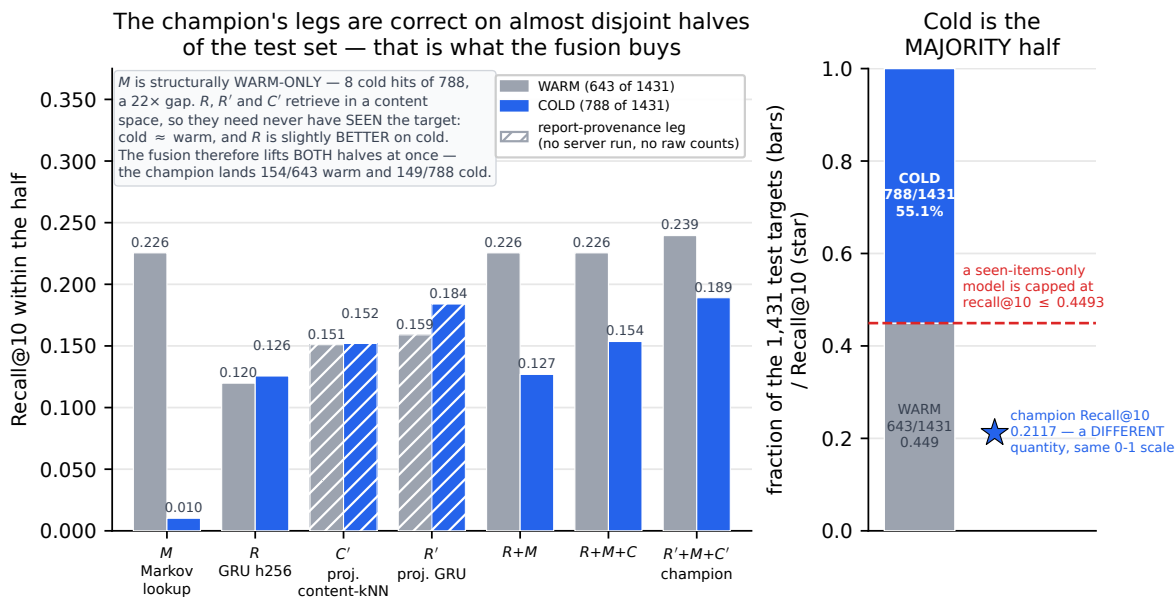


Figure 12: The one decomposition that explains the champion. *Left*: Recall@10 split into warm and cold halves for each champion leg and each blend stage. The Markov leg collapses to 0.010 on cold targets (8 of 788) against 0.226 warm, whereas the projected content leg reads cold 0.152 ≈ warm 0.151 and the base GRU is fractionally *better* on cold than warm — so the fusion lifts both halves at once (154/643 warm, 149/788 cold). The *R'* and *C'* bars are hatched because those standalone figures are report provenance from an off-server driver, with no server run and no raw hit counts. *Right*: the split’s composition — the warm fraction *is* the seen-items-only ceiling, $\text{Recall@10} \leq 0.4493$, against $788/1431 = 55.1\%$ cold. Any model that cannot retrieve an unseen item is arguing over the left-hand 45% only.

7 The learned projection wins because it rebuilds the retrieval metric

Rationale. Before the projection, both the GRU leg and the content-kNN leg retrieved by cosine in the same unsupervised PCA-192 space. That space is variance-ordered and completely target-agnostic: it was fit to *reconstruct* the item feature matrix, so its dominant directions are whatever has the most raw variance — on this corpus loudness, era and popularity — and not “tracks that follow each other”. The top ten principal components carry 65 % of the variance with essentially zero text loading. The projection replaces the *metric* rather than the model.

What is actually learned. `seq_blend.fit_projection` learns a single square matrix W (192×192 , no bias, initialized to the identity; Adam, lr 1×10^{-3} , 40 epochs, batch 512) from *train* consecutive item pairs (u, v) with self-loops dropped. The objective is the in-batch InfoNCE of Section 2 at $\tau = 0.07$: for a batch of 512 pairs, embed $a_i = \text{normalize}(Wz_{u_i})$ and $p_j = \text{normalize}(Wz_{v_j})$ and require each item’s true successor to outscore the 511 other rows’ successors.

What W is, stated exactly. Up to the row normalization, ranking by $\langle Wz_u, Wz_v \rangle = z_u^\top (W^\top W) z_v$ means the projection is a **learned Mahalanobis metric** on the original 192 dimensions, with positive-semidefinite Gram matrix $W^\top W$. Two things follow immediately. First, by the singular value decomposition $W = U\Sigma V^\top$ a general square W is a rotation, then a per-axis rescaling, then *another* rotation — so it can both re-orient the axes along which “followed by” varies and rescale them, which is more than a diagonal reweighting of the existing PCA coordinates could do. Second, $z_u^\top (W^\top W) z_v$ is *symmetric in u and v by construction*: the object being learned is a “these two go together” relation, not a directed “ u is followed by v ” one. That is exactly why the directional two-tower ablation below is the right test of whether the symmetry is a limitation, and why its losing is informative rather than incidental. **The honest gap:** the spectrum of $W^\top W$ has never been inspected on this corpus, so the natural next sentence — *which* directions were stretched and which shrunk — is not something this document can assert. What is measured is only the consequence: after the fit, cosine in the new space behaves like transition affinity. Two of the three legs are defined by cosine in that space, so both improve at once; the Markov leg is item-ID based and is untouched by construction. And because W is a function of *content only*, an item that never appeared in training still has a position in the projected space — this is the one channel with cold reach, and the cold half is 55 % of the test set. The supervision is leak-free: train pairs and intrinsic latents only.

Results. $R' + M + C'$ reaches Recall@10 0.2117 [0.191, 0.233] at MRR 0.1220, paired Δ over the $R + M + C$ crown $+0.0259$ [$+0.012$, $+0.041$], CI entirely above zero. Both shared-metric legs improve and the untouched leg does not: content-kNN $C \rightarrow C'$ goes $0.0685 \rightarrow 0.1516$ — a $2.2\times$ lift on a leg with *no trainable parameters of its own* (the kNN rule is parameter-free; the 36 864 parameters of the shared W it now retrieves inside are counted once, in the champion’s total), so the entire change is the geometry — the GRU leg $R \rightarrow R'$ goes $0.1230 \rightarrow 0.1719$ to 0.1761 across seeds, and Markov stays at 0.1069. Unlike the deterministic GRU path this result has genuine seed variance (random init, shuffled batches): over seeds 1337/7/99 the blend reads 0.2117/0.2110/0.2145 with paired Δ $+0.026/+0.025/+0.029$, every CI above zero. It is then hardened on a split disjoint in both index and time (`seq-20260718-211238`, the earlier chronological band, byte-identical item space, cold rate 0.64): 0.1705 against the crown’s 0.1279 and $R + M$ ’s 0.1174, paired Δ $+0.0426$ [$+0.0280$, $+0.0573$] — a *wider* margin on the colder split, exactly what the mechanism predicts, since when more targets are cold the cold-reaching leg carries proportionally more of the win. Figure 13 shows both splits.

The axis ablations, and their status. Four cells sharpen what is doing the work, and none of them carries a paired CI — they are *nominal single-run* comparisons and are labelled as such here and in Table 2. InfoNCE $\gg$ BPR: replacing the softmax-over-in-batch- negatives objective with pairwise BPR — Bayesian Personalized Ranking, the standard pairwise recommender loss, which for each positive samples *one* negative and maximizes $\log \sigma(s_{\text{pos}} - s_{\text{neg}})$ — collapses the blend to 0.1579 and 0.1523, roughly 0.055 below the default and *below even the unprojected crown*, so the specific shape of the contrastive loss is load-bearing and not merely “some pairwise supervision”. Full rank $\approx$ rank 64: constraining W to 64×192 gives 0.2055 against 0.2117, a nominal -0.006 and a tie inside the ± 0.015 band — *the gain is the supervision, not the capacity*. Temperature is mildly monotone over the tested range ($0.05 \rightarrow 0.1929$, $0.07 \rightarrow 0.2117$, $0.10 \rightarrow 0.2229$) and $\tau 0.10$ ’s $+0.0112$ sits inside the band, so the decision rule did not re-promote it. A directional two-tower W_q/W_k variant — which drops the $W^\top W$ symmetry by giving u and v separate maps — scored C' 0.1349 against the symmetric single W ’s 0.1516, so the symmetry is not the binding constraint, and a symmetric “these two go together” relation is what this corpus supports.

Interpretation — the nuance that is easy to state wrongly. A separate campaign asked whether the frozen equal-weighted PCA-192 vector is itself the bottleneck, building eight alternative 192-dimensional item spaces under parity gates and evaluating each under two readouts. Under the *frozen* GRU readout, *six of eight* beat it with paired CI above zero (0.0252 to 0.0356), so at the raw level the PCA-192 geometry genuinely is a handicap. Under the *champion* readout, with the projection re-fit on each space, *not one* variant beats it: the two nominally positive spaces straddle zero and two are significantly worse. The correct reading is that pre-whitening the input and the learned projection are the *same fix applied in two places*, so they do not stack — a full-rank linear map over the same 192 dimensions can rescale and rotate away the loud inert directions, and InfoNCE gives it the supervision to know *which* ones. It is emphatically not “the representation does not matter” (it demonstrably does, six times over with CI above zero) and not “whitening helps the champion” (it does not). A sub-finding sharpens it: dropping the numeric and acoustic blocks *helps* the frozen GRU ($+0.0335$) but *hurts* the champion (-0.0168), so the lesson is not “drop the loud blocks” but “let a supervised map decide their weight”. The honest caveat is that this cross-readout conclusion rests on a *null* on the champion readout, which is weaker evidence than a win. Because those eight spaces are dataset variants rather than models they are otherwise out of scope here.

Sources: 2026-07-18-nexttrack-literature-fit-campaign.md, 2026-07-18b-nexttrack-projection-registration-and-scan.md, and 2026-07-18c-nexttrack-representation-exploration.md.

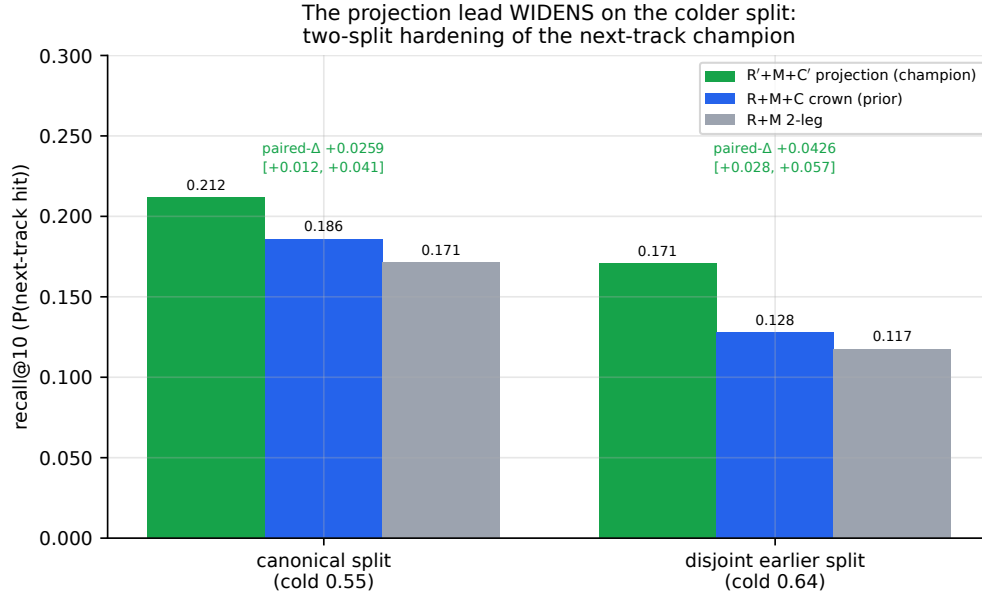


Figure 13: The champion’s lead *widens* on the colder split, which is the mechanism’s own prediction. Two splits $\times$ three blends: the canonical split (cold rate 0.55) and a disjoint earlier-holdout split (cold rate 0.64), with $R'+M+C'$ in green (0.2117 / 0.1705), the unprojected $R+M+C$ crown in blue (0.1859 / 0.1279) and the two-leg $R+M$ in gray (0.1712 / 0.1174). The annotated paired deltas are +0.0259 [+0.012, +0.041] and +0.0426 [+0.0280, +0.0573], both CIs above zero. Absolute recall is lower on the right-hand split by construction; the verdict is the paired Δ , not the level. *Sources: 2026-07-18-nexttrack-literature-fit-campaign.md, and 2026-07-18b-nexttrack-projection-registration-and-scan.md.*

8 Recurrence beats pooling because the pooled summaries are already inside the state

Rationale. `seq-ann` is built specifically as the GRU’s non-recurrent twin, and the comparison is unusually clean because everything except the architecture is reused verbatim: `seq_ann.py` imports `make_batches` and `step_loss` from `seq_nexttrack`, so objective, batching and early stopping are byte-for-byte identical, and the live API confirms matched hyperparameters ($h = 256$, InfoNCE, 40 epochs, lr 1×10^{-3} , dropout 0.1, seed 1337). Where the GRU carries a hidden state updated by learned gates, the ANN builds a fixed-width bag-of-prefix at each step — the concatenation of the *mean* of the prefix latents so far and the *last* item’s latent — and feeds it through a ReLU MLP. Both are strictly causal, so neither leaks. The difference is therefore exactly: a mean plus a special-cased most-recent item, versus a state whose gates decide per dimension and per step how much of the incoming item to write and how much context to forget. A mean is permutation-invariant and weights a track from twenty steps ago as much as the one just played; a gated state can hold a stylistic thread through an interruption and can represent “A then B” differently from “B then A”.

Results. Head-to-head on the same PCA-128 space `seq-20260715-030509`: ANN 0.09015 (129 of 1431) at MRR 0.03941 against the GRU anchor’s 0.11461 (164) at MRR 0.0489, paired $\Delta -0.0245$ $[-0.039, -0.011]$, CI entirely below zero; the LSTM on the same space reaches 0.12718 (182), so the pooled model is beaten by both recurrent cells. Where it loses is informative and was recomputed here from the same predictions pipeline: ANN warm 0.1026 (66/643) / cold 0.0799 (63/788) against the GRU’s 0.1182 / 0.1117, i.e. it gives up 10 warm hits but 25 cold ones — the damage concentrates on the cold half. *Why* it concentrates there is a conjecture this document cannot settle, and it must not be stated as the earlier Markov-versus-GRU argument was: neither the ANN nor the GRU has any item-memory mechanism, so “you cannot lean on having seen the item” does not distinguish them — both are pure content retrieval on both halves. The plausible reading is about *density*: warm targets are by definition items the user played before, so they sit in the well-trodden, densely covered regions of the latent space where a sloppy predicted point still lands one of ten neighbours, whereas cold targets sit in sparse regions where the precision of the point is what decides. That is untested here — no per-region density measurement exists — and it is offered as the likely mechanism, not a result. Hit-set overlap: 92 of the ANN’s 129 hits are also GRU hits and only 37 are ANN-only, so the pooled model is close to a strict subset — which is also why it is redundant as a blend leg (Section 6). These warm/cold and overlap figures are single-run, uncertainty-unquantified counts.

Interpretation — two triangulations and one reverse test. First, the explicit pooled summaries are drastically weaker *even with recurrence over them*: a two-tower arm reading only the running prefix mean scores 0.06639 (95) and one reading only step-to-step differences 0.05800 (83), against 0.12299 for a single GRU over raw latents. Both of those ablation arms carry the harmful `fusion_layers = 1` head (Section 10.3), so part of the ~ 0.057 gap is the head rather than the view; the clean within-regime contrast is against the latent/latent arm at the *same* depth, 0.06639 against 0.10273, a still-decisive -0.036 . The direction of the finding survives; its magnitude does not. Second, handing a raw-latent tower an *explicit* running mean or difference adds nothing — latent/cummean versus latent/latent is -0.00559 $[-0.01747, +0.00699]$ and latent/delta is -0.01118 $[-0.02306, +0.00070]$, both straddling zero — because a GRU’s update gate is a leaky accumulator (a running mean is inside its hypothesis space) and its candidate/gate interaction covers first differences. An interpretability read on that arm is consistent with the derived views being lossy functions of what the raw-latent tower already held: the two towers’ representation geometries

are only partly distinct (CKA 0.458 — a *middling* similarity, so there is real new linear structure between them) yet the sparse-autoencoder decomposition recovers *zero* next-item concepts in the second tower that the first does not already carry. New geometry, no new concepts. Third, the reverse test shows the fusion machinery is not broken: pairing the two *individually hopeless* views with each other beats both parents with CI above zero (+0.02865 and +0.02027) at a hopeless absolute 0.08665. Two-view co-adaptation is a real mechanism; it simply has nothing left to contribute once a raw-latent tower is present. Figure 14 draws the argument. Capacity is separately ruled out, with the caveat of Section 10.5: on PCA-192 single-GRU width is flat at the endpoints, $h = 425$ at $2.2\times$ the parameters tying $h = 256$ (-0.00210 [$-0.01118, +0.00629$]) and $h = 454$ at $2.5\times$ also tying, so the ~ 0.025 recurrence gap cannot be bought with parameters. And the GRU’s advantage *widens* multi-step: on three-step continuation ($n = 1172$) it beats the first-order bigram on both suffix_recall@10 (0.0963 versus 0.0869) and cont_prec@10 (0.0273 versus 0.0236).

The honest weakness. The ANN datapoint is a *single unreplicated run on PCA-128*. It has never been measured on the canonical PCA-192 split, so it must not be read in the same column as the 0.12299 bar (Table 2, dagger), and the only defensible GRU-versus-ANN statement is the same-space one above.

Sources: 2026-07-15c-nexttrack-phase2-model-sweep.md, 2026-07-25-nexttrack-dual-tower-fusion-scan.md, and 2026-07-26-nexttrack-pre-encoder-scan.md.

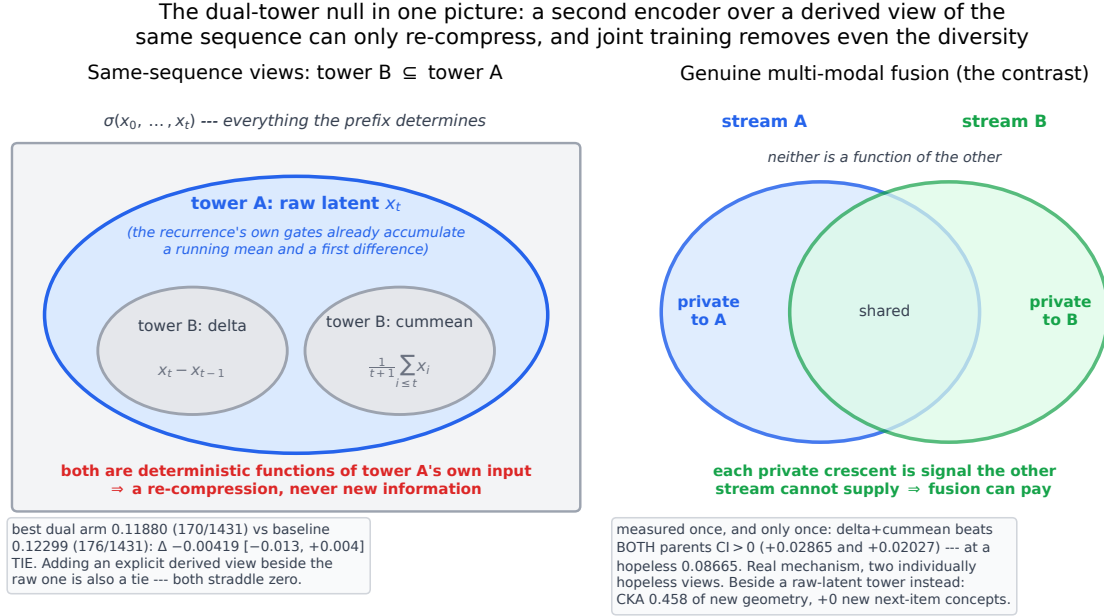


Figure 14: Why a second tower cannot add information here. *Left*: the three tower views — latent x_t , delta $x_t - x_{t-1}$, cummean the running prefix centroid — are all deterministic, strictly causal functions of the *same* prefix, so $\sigma(\text{tower B}) \subseteq \sigma(\text{tower A})$: tower B holds a re-compression of what tower A already receives, never new information, and the recurrence’s own gates already accumulate a running mean and a first difference. Measured, the best dual arm 0.11880 (170 of 1431) ties the 0.12299 (176) baseline, paired $\Delta -0.00419 [-0.013, +0.004]$. *Right*: what genuine multi-modal fusion looks like, where each stream has a private region the other cannot supply — measured exactly once on this corpus, with **delta+cummean** beating *both* parents with CI above zero (+0.02865, +0.02027) at a hopeless absolute 0.08665. The inner ellipses are drawn strictly inside tower A’s boundary because the panel asserts a subset relation. The right panel’s “CKA 0.458” is a similarity, not a fraction of newness: it says the two towers’ representation geometries are only *partly* alike, which makes the accompanying “+0 new next-item concepts” the informative half of the pair. All the recall figures quoted on this figure are **fusion_layers** = 0 arms except the parents of the **delta+cummean** contrast, which are **fusion_layers** = 1 throughout and therefore internally comparable.

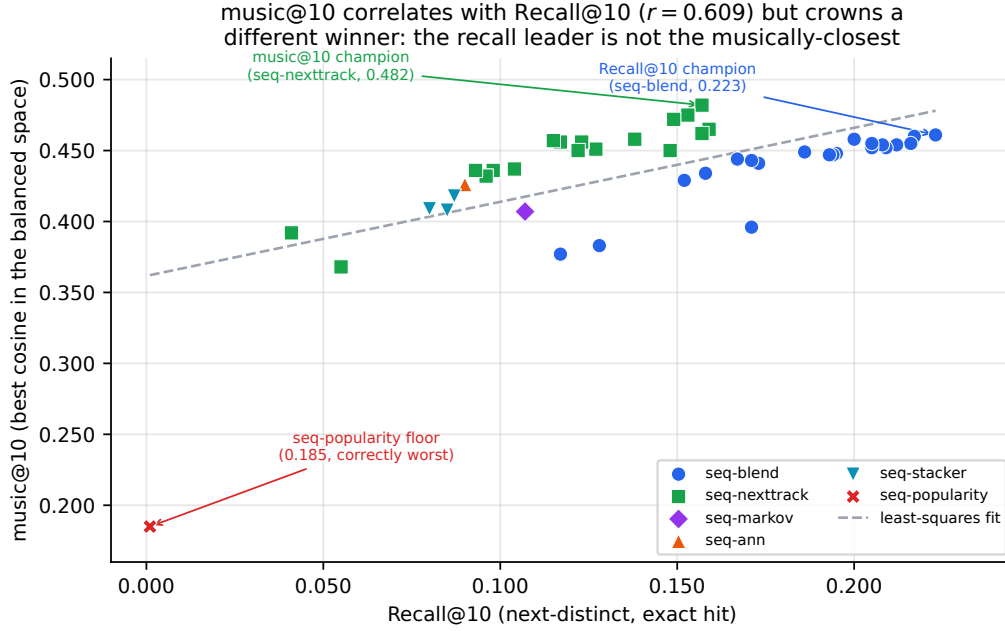


Figure 15: The families differ in *kind*, not only in rank. music@10 against Recall@10 for all 53 next-track runs, one marker per run, coloured and shaped by predictor family: **seq-blend** (blue circles), **seq-nexttrack** (green squares), **seq-markov** (purple diamond), **seq-ann** (orange triangle), **seq-stacker** (cyan down-triangles), **seq-popularity** (red $\times$ at the bottom left). The two metrics correlate at $r = 0.609$ but crown different winners: the content-regressing **seq-nexttrack** family lands the *musically closest misses* (best music@10 0.482) while the blend family wins exact recall. **One in-figure annotation is stale and must be read against Section 5:** the point it labels “Recall@10 champion (0.223)” is the τ 0.10 scan cell whose nominal edge was judged sub-noise and whose default was held — it is the highest Recall@10 measured on this split, but the champion is $R' + M + C'$ at 0.21174. Read against Section 3, this is why music@10 makes a usable grounding factor: it is a smooth superset of recall, not a substitute for it. *Source: 2026-07-20-musical-distance-metric.md.*

9 MMR is the first honest crown lever, and it de-concentrates artists for free

Rationale. Every prior attempt to raise `holisticness@10` raised it by predicting worse (Section 3). MMR re-ranking is different for a structural reason: it is eval-time only, so the trained weights are bit-identical across λ arms and $\lambda = 1$ is an exact identity gate, and its single greedy penalty moves *three* of the crown’s four factors the same way at once. `ild` is literally the quantity the penalty maximizes, so it rises by construction. Eagerness falls as a *side effect* with no notion of artist anywhere in the rule, because same-artist tracks are sonic near-neighbours and the max-similarity penalty preferentially evicts them. And `music@10` *rises*, because a sonically spread top-ten covers more of the neighbourhood around the truth, so the best-of-ten cosine to the true next track goes up. Only exact recall pays. Figure 16 draws the geometry; Figure 18 shows the original offline sweep.

Results. Ten arms on the canonical split, same host, serialized, none failed; every arm’s H recomputed from `predictions.json` under one code path and validated against storage to $|\Delta| \leq 2.1 \times 10^{-7}$; all statistics paired bootstrap with all five facets and hence H recomputed for both arms on each resampled index set. **Eight of eight $\lambda < 1$ arms lift H with paired ΔH CI entirely above zero** (Table 3). The best *free* operating point is the single GRU at $\lambda = 0.7$ with `mmr_pool` 50: H 0.030355, +22.1% over the all-time healthy bar 0.024872 and 5.4 measured half-widths above it, at Recall@10 0.120196 (172) with Δ recall -0.002908 $[-0.008386, +0.002795]$ *straddling* zero and $|\Delta|$ far inside the ± 0.015 band. It also beats the nearest pure- λ point on H by $+0.000889$ $[+0.000497, +0.001360]$ at a recall tie, so its rank as best free arm is resolved rather than nominal.

Results — the facets, with the arm named. The three-factors-together move is clearest on the GRU’s deepest λ arm, **B3** ($\lambda = 0.7$, **pool** 200), which is *priced* rather than free: there, as λ goes $1.0 \rightarrow 0.7$, `ild` rises $0.4536 \rightarrow 0.5314$, `artist_adj` falls $0.3474 \rightarrow 0.3129$, `artist_conc` falls $0.3559 \rightarrow 0.2707$ and `music@10` *rises* $0.4558 \rightarrow 0.4747$, with `mood_coh` moving only $0.4788 \rightarrow 0.4585$ (-4.2%). The *free* arm B4 (same λ , pool 50) makes the same moves less far — `ild` 0.5017 , `artist_adj` 0.3270 , `artist_conc` 0.3019 , `music@10` 0.4670 , `mood_coh` 0.4671 — which is precisely the trade the pool axis buys back. Every quoted facet is a cosine in the content metric M (Section 2), which is also the space MMR’s penalty acts in; that shared space is why one knob moves four numbers. The instrument transferred exactly: an offline sweep on the champion checkpoint had projected $0.2103/0.1936/0.1642/0.1097$ for λ $0.9/0.7/0.5/0.3$ and the native runs returned $0.21034/0.19357/0.16422/0.10971$.

Interpretation — the mechanism, resolved. A pre-registered question asked whether artist concentration is *responsive* to a sonic penalty or *sticky*. It resolves unanimously to responsive: in eight of eight arms $|\Delta \text{artist_conc}|$ exceeds $|\Delta \text{artist_adj}|$, by 1.59 to $3.06\times$, where the sticky hypothesis required a ratio below 0.5. So a rule with no notion of artist in it de-concentrates artists faster than it reduces seed echo, and the planned explicit artist penalty is consequently dropped as the next move. The ratio also *shrinks* monotonically as λ falls ($3.06 \rightarrow 1.59$ on the champion), i.e. the easy de-concentration is bought first and the mechanism saturates. Because the crown takes $\max(\text{artist_adj}, \text{artist_conc})$ and concentration falls $\sim 2.5\times$ faster, the two must cross: past that switch the eagerness factor becomes the stiff one and each further recall point buys progressively less crown. Note that this verdict rests on the *sign and ratio* of point estimates across eight arms — its strength is unanimity, not an interval on the ratio. The ΔH and Δ recall figures, and hence the tiers, are paired-CI-backed.

Why the plain GRU benefits and the champion does not. The pre-registered expectation was

Table 3: The eval-time MMR λ frontier, ten arms on seq-20260715-131139 (1431 test sessions), grouped by anchor and sorted within group by λ . λ and `mmr_pool` are *eval-time* parameters: within a group the trained weights are bit-identical, so $\lambda = 1.0$ is an exact identity control. Paired Δ is a per-session paired bootstrap against that control (2000 resamples, rng 1337); CI bounds are given to three decimals, and a CI straddling zero is a *tie*. H is the *generation-3 (live)* composite. Tiers are the campaign’s own decision rule: TIER-1 *free* (ΔH above zero at a recall tie), TIER-2 *priced* (Δ recall CI below zero), TIER-3 *rejected* (frontier mapping only). **Six definitions were registered on 2026-07-26 and none was promoted**; the recall champion is unchanged. **Bolding.** The bolded row is B4, the best *free* operating point, and the two bolded cells are the best H and ΔH among the free arms. **Recall is deliberately unbolded in this table:** it is not a maximization target here but a *constraint* — each group’s $\lambda = 1.0$ control is the highest recall in its group by construction, since MMR can only reorder a list the control already produced, and a free arm’s merit is that its recall Δ *straddles* zero rather than that it is large. Recall also must not be compared *across* the two groups: the anchors are different models. *Source: 2026-07-26-nexttrack-mmr-lambda-frontier.md.*

Arm	λ / pool	R@10 (raw)	Δ recall [95% CI]	H gen 3	ΔH [95% CI]	Tier
B0 GRU h256	1.0 / —	0.12299 (176)	—	0.024538	—	control
B1 GRU h256	0.9 / 200	0.12089 (173)	−0.0021 [−.006, +.002]	0.026761	+0.0022 [+0.0016, +.0029]	TIER-1 free
B2 GRU h256	0.8 / 200	0.11950 (171)	−0.0036 [−.009, +.001]	0.029458	+0.0049 [+0.0040, +.0059]	TIER-1 free
B4 GRU h256	0.7 / 50	0.12020 (172)	−0.0029 [−.008, +.003]	0.030355	+0.0058 [+0.0048, +.0068]	TIER-1 free
B3 GRU h256	0.7 / 200	0.11461 (164)	−0.0085 [−.016, −.001]	0.034171	+0.0096 [+0.0083, +.0110]	TIER-2 priced
A0 $R'+M+C'$	1.0 / —	0.21174 (303)	—	0.008603	—	control
A1 $R'+M+C'$	0.9 / 200	0.21034 (301)	−0.0013 [−.008, +.005]	0.010435	+0.0018 [+0.0015, +.0022]	TIER-1 free
A2 $R'+M+C'$	0.7 / 200	0.19357 (277)	−0.0181 [−.029, −.008]	0.015289	+0.0067 [+0.0057, +.0077]	TIER-2 priced
A3 $R'+M+C'$	0.5 / 200	0.16422 (235)	−0.0473 [−.061, −.033]	0.023576	+0.0150 [+0.0132, +.0168]	TIER-3 rejected
A4 $R'+M+C'$	0.3 / 200	0.10971 (157)	−0.1019 [−.120, −.084]	0.031626	+0.0230 [+0.0206, +.0256]	TIER-3 rejected

the opposite: the champion’s eagerness is $1.9\times$ the GRU’s, so its $(1 - \text{eager})$ factor enters at 0.308 against 0.644 and it has far more headroom. The measurement says no — at $\lambda = 0.7$ the GRU reaches H 0.034171 against the champion’s 0.015289, a $2.24\times$ gap whose *absolute* size grows as λ falls (0.01633 at $\lambda = 0.9$, 0.01888 at $\lambda = 0.7$) even though the ratio itself narrows slightly, from $2.56\times$ to $2.24\times$, over the only two λ values where both anchors were run (Figure 17). The reason is in the facets. At *matched* $\lambda = 0.7$ and pool 200, the champion’s `mood_coh` falls 0.3982→0.3643 (−8.5%) against the GRU’s 0.4788→0.4585 (−4.2%) — a factor of two, not a collapse. Pushed to $\lambda = 0.3$ the champion’s `mood_coh` does collapse to 0.2378 (−40%), but no GRU arm was run that deep, so the often-quoted 40%-versus-4% contrast is *not* like-for-like and the honest matched figure is the $2\times$ one. The champion’s recall is built substantially on artist adjacency supplied by its Markov leg, so diversifying away from the seed artist also walks it out of the prefix’s mood neighbourhood — it pays twice. MMR’s value therefore scales with the *health* of the other three factors, not with headroom to de-eager. Figure 19 shows the same asymmetry from the inside, as the geometry of the candidate set the two models produce.

What is not established. Nothing here is promoted. Six definitions were *registered* on 2026-07-26; a mandatory second-split robustness confirm and user approval are still pending; the best free arm’s rank over its neighbour is resolved at only ~ 0.9 half-widths of its own comparison, on one split. And the campaign leaves a live calibration question open: the grounding floor 0.34 was chosen to *cancel* recall-destroying H gains, and MMR is the first lever that moves music@10 the *other* way, so the constant is currently operating outside the regime it was calibrated for.

Sources: 2026-07-22-session-holisticness-and-antieager-levers.md, and 2026-07-

Why sonic diversification de-eagers artists for free: same-artist tracks are sonic near-neighbours, so the max-similarity penalty evicts them first (the H1 verdict)

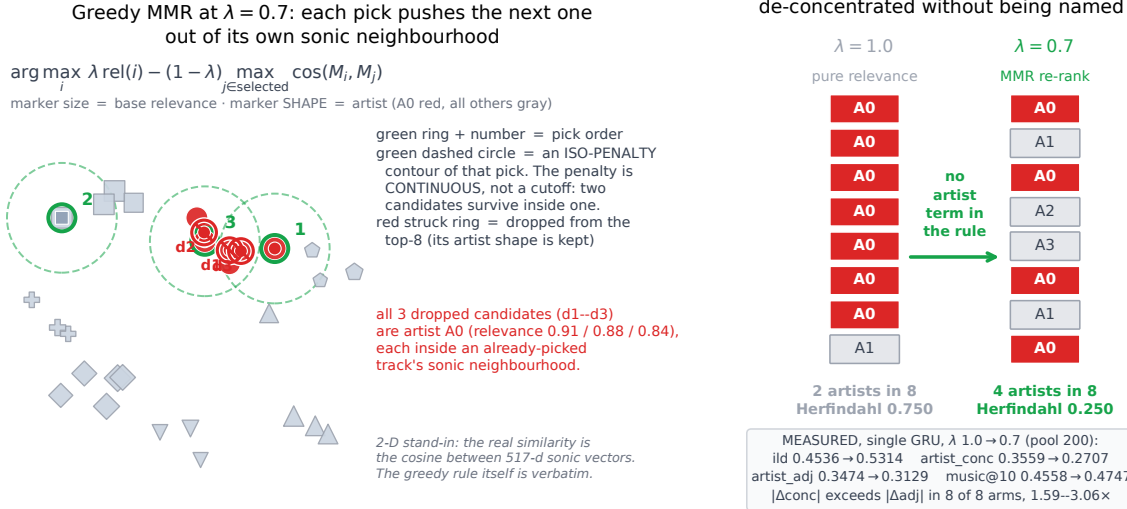


Figure 16: Sonic diversification de-eagers artists for free. *Left*: the greedy rule $\arg \max_i \lambda \text{rel}(i) - (1 - \lambda) \max_{j \in \text{selected}} \cos(M_i, M_j)$ applied at $\lambda = 0.7$ over a candidate pool. Each candidate keeps its own artist marker *shape*; a red ring marks the three candidates the re-rank drops from the top eight, and all three are the *same* artist as an already-picked track sitting close to it. Each pick's green ring is an *iso-penalty contour*, not a cutoff: the penalty $\max_j \cos(M_i, M_j)$ is continuous, so a sufficiently relevant candidate survives inside a contour — two do, which is the mechanism working correctly rather than an inconsistency. *Right*: the two resulting lists — pure relevance returns seven of eight from one artist, MMR returns four artists — with no artist term anywhere in the rule. The pool is schematic (real similarity is cosine between 517-dimensional vectors and is not drawable in two dimensions, so planar proximity stands in) and the list's own Herfindahl arithmetic is the illustration's; the *measured* facet moves sit in the separate callout, and they are arm B3 (single GRU, `mmr_pool` 200, TIER-2 *priced*), not the free arm: at $\lambda 1.0 \rightarrow 0.7$, ild 0.4536 $\rightarrow$ 0.5314, artist_conc 0.3559 $\rightarrow$ 0.2707, artist_adj 0.3474 $\rightarrow$ 0.3129, music@10 0.4558 $\rightarrow$ 0.4747, with $|\Delta \text{conc}| > |\Delta \text{adj}|$ in eight of eight arms at 1.59 to 3.06x.

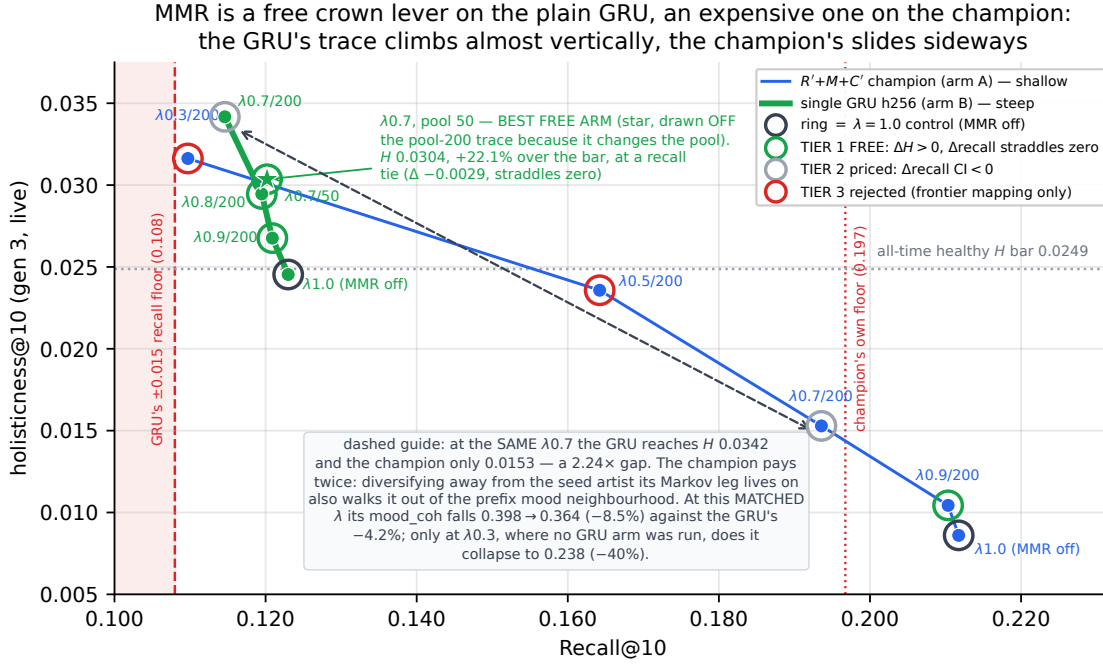


Figure 17: The same lever is free on one model and ruinous on the other. The MMR λ frontier in (Recall@10, generation-3 H) space, one traced curve per anchor at fixed `mmr_pool` 200: the single GRU's thick trace climbs almost *vertically* (H 0.0245 $\rightarrow$ 0.0342 over λ 1.0 $\rightarrow$ 0.7) while the champion's slides *sideways*, spending -0.102 recall to buy $+0.023$ H . Note that the GRU trace's top end, H 0.0342, is arm B3 — TIER-2 *priced*, Δ recall CI below zero — so the vertical climb runs one step past the free region; the star is the genuinely free best arm B4 and it is drawn *off* the trace because it changes `mmr_pool` to 50 and so does not belong to the same fixed-pool sweep. Tier is encoded as a ring (free / priced / rejected / $\lambda=1$ control), the λ labels carry their pool where it differs, and both anchors' own ± 0.015 recall floors are marked (0.108 for the GRU, 0.197 for the champion). The dashed double arrow is an eye guide between the two anchors at matched $\lambda = 0.7$, not a fitted curve. **Nothing plotted here is promoted.**

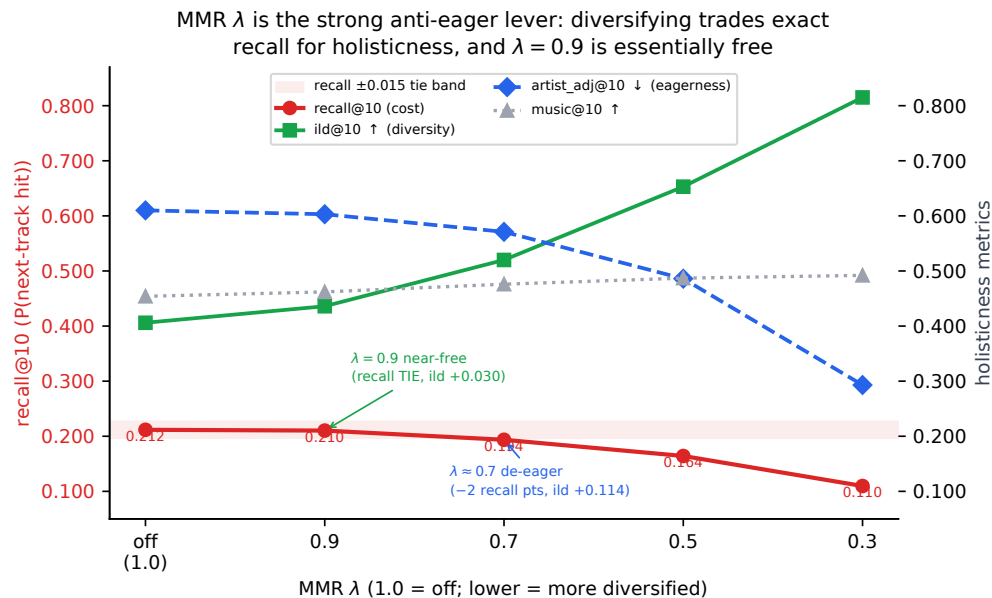


Figure 18: The original offline λ sweep on the champion checkpoint, which is what motivated running the native arms of Table 3. Left axis: Recall@10 (red) falling 0.2117→0.2103→0.1936→0.1642→0.1097 across λ off/0.9/0.7/0.5/0.3, with the ± 0.015 tie band shaded around the λ -off point. Right axis: ild@10 rising 0.406→0.815, artist_adj@10 falling 0.610→0.293, music@10 rising 0.454→0.492 — the three-factors-together move, including the grounding factor, that distinguishes this lever from the degenerate ones. **The right axis carries the individual facets, each on its own 0–1 scale, not H itself** — the figure’s own axis label calls them “holistiness metrics”, which is the facet suite, and H (~ 0.01 – 0.03 here) is nowhere on this plot. The transfer to the native runs was exact to four decimals. *Source: 2026-07-22-session-holistiness-and-antieager-levers.md.*

The eigen-shape of a next-track belief: the champion blend's candidate cloud collapses onto ONE axis (leading share $0.591 \rightarrow 0.704 \rightarrow 0.897 \rightarrow 0.972$ over four steps) where the single GRU shows no trend (AUTHOR-RUN, SINGLE SEED, ONE PREFIX — not a campaign result and not in the experiment record)

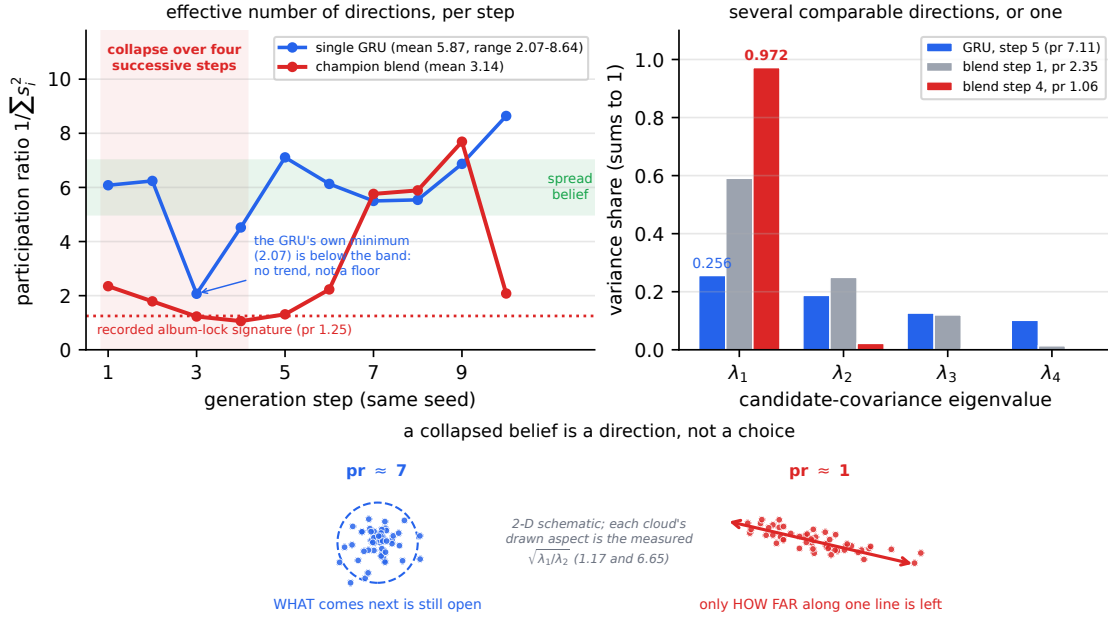


Figure 19: The eigen-shape of a next-track belief, and why the champion is the harder model to diversify. **Provenance first, because it is unlike every other figure here: this panel is not in the record.** It is two **extend** invocations run by the author for this document, on one five-item seed prefix, single seed, ten *generation steps* — a generation step being one round of “rank, take the top candidate, append it to the prefix, re-rank”, i.e. the autoregressive continuation mode rather than the single-step protocol of Figure 1. No server run record exists for either invocation and no campaign report contains these numbers; they are reproducible only from `docs/figures/make_concept_figures.py`. Read it as an illustration of a mechanism, never as evidence. *Left*: the participation ratio $1/\sum_i s_i^2$ (Section 2) of the candidate-set variance spectrum at each step. The champion blend falls *monotonically* over four successive steps (pr $2.35 \rightarrow 1.79 \rightarrow 1.23 \rightarrow 1.06$, mean over all ten 3.14) while the single GRU fluctuates non-monotonically without trend over the whole run (2.07 to 8.64, mean 5.87) — note that the GRU’s own minimum, 2.07 at step 3, falls *below* the band drawn for it, so the contrast is monotone-collapse versus no-trend, not inside-band versus outside. *Middle*: three spectra — the GRU at its step-5 point (pr = 7.11, leading share 0.256) against the blend’s first and fourth steps, the latter putting 0.972 of the variance on a single axis. *Right*: what that means geometrically — a spread cloud leaves *what* comes next open, a collapsed one leaves only *how far* along one line; each cloud’s drawn aspect is the measured $\sqrt{\lambda_1/\lambda_2}$. The one external check available is that the blend’s collapsed step reproduces the album-lock signature recorded in `predictors/seq_extend.py` (pr 1.23 versus 1.25, leading share 0.897 versus 0.89).

10 The failures, and the measured mechanism in each case

Five design axes have been refuted on this corpus. What makes them worth a section is not the nulls themselves — Section 11 shows the instrument could not have detected a moderate win anyway — but that in four of the five the *mechanism* was subsequently measured rather than argued.

10.1 Dual-tower view fusion: the second tower’s input is measurable with respect to the first’s

Rationale and mechanism. The failure can be stated before any run. The three tower views are all deterministic, strictly causal functions of the *same* prefix, so $\sigma(\text{view}_t) \subseteq \sigma(x_0..x_t)$: tower B holds a lossier summary of what tower A already receives and can add optimization-level diversity, never new information. The second half of the mechanism is about the ensemble story: joint end-to-end training is precisely what *destroys* the independence that makes combination pay. The fixed z-blend works because its legs are trained separately and never see each other’s gradients; two towers trained jointly through one loss co-adapt into a single estimator with two halves. That is why the same corpus supports a 0.212 blend and refuses a 0.119 joint dual tower.

Results. Zero of the eight arms in the 2026-07-25 dual-tower scan beat the baseline with paired CI above zero. The best of those eight, latent/latent at `fusion_layers = 0`, reaches 0.11880 (170) against 0.12299 (176), -0.00419 $[-0.01258, +0.00419]$, a tie. (One *later* dual arm, T1-f0 in the 2026-07-26 batch, nominally exceeds the bar at 0.12369; it also ties, $+0.00070$ $[-0.01188, +0.01328]$, and Section 10.2 explains why it is a repair rather than a win. No dual arm anywhere clears zero.) Capacity is ruled out by the single-GRU width control at 871 017 parameters, which ties the bar at 0.098 % parity with the `fusion_layers = 1` dual arms (871 872) — note that the parity is with those, not with the 789 696-parameter *f0* arm just quoted, which the control outweighs by 10 % and still only ties. Figure 20 gives the `fusion_layers = 1` view matrix. The view attribution rule makes the mechanism decidable: mixed cells containing a raw-latent tower *tie* their latent/latent parent and beat only their weaker parent, whereas the one pair with no raw-latent tower at all beats *both* its parents with CI above zero. Complementarity is real and measured; it has nothing left to contribute once a raw-latent tower is present.

A citation that could not be made. Two further diagnostics were proposed for this section — a rank comparison of the concatenated two-tower representation against each tower alone, and a cross-prediction R^2 between identical-view towers — and neither exists anywhere in this instance: not in any `experiments/*.md` report, not in `PROJECT-FACTS.md`, not in any run’s metrics. Under this project’s integrity rules the figures are omitted and their numbers are not printed here even as illustration. The citable substitute for the diversity-collapse claim is that two independently initialized *same-view* towers fused jointly tie both the baseline and the parameter-matched control, buying none of the implicit-ensemble gain that independently trained legs demonstrably do buy.

Source: 2026-07-25-*nexttrack-dual-tower-fusion-scan.md*.

10.2 The MLP pre-encoder: a rectifier deletes half of a zero-centred latent

Rationale. This is the sharpest mechanism on record because the design left the ReLU as the only variable. A GRU’s own input transform is already linear: each gate computes $W_i x$. Prepend a linear pre-encoder V of shape $(\text{pre_hidden} \times 192)$ and the composition is $W_i(Vx) = (W_i V)x$; when $\text{pre_hidden} \geq \text{latent_dim}$, V is generically full rank, so $W_i V$ ranges over exactly the same space of matrices as W_i . **A linear pre-encoder is therefore provably expressivity-neutral** — it

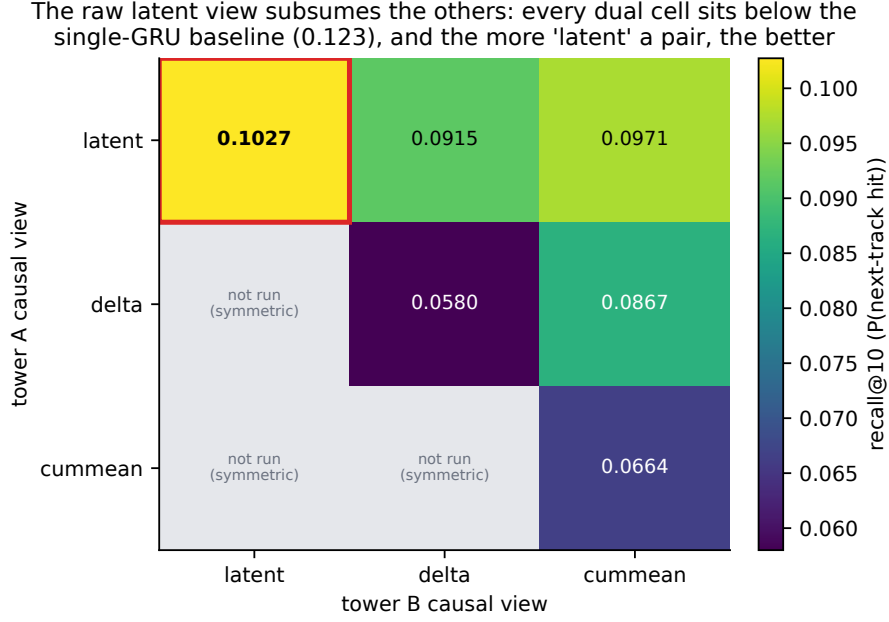


Figure 20: The raw latent view subsumes the engineered ones. Recall@10 for the two-tower view pairing over $\{\text{latent}, \text{delta}, \text{cummean}\}$, **all six cells at fusion_layers = 1**, one run each (upper triangle only; the grey lower cells were not run because the pairing is symmetric). Reading the diagonal is the argument: latent/latent 0.10273, delta/delta 0.05800, cummean/cummean 0.06639 — the views that discard raw item identity are catastrophic even with a full recurrence over them. **Every cell sits below the single-GRU baseline 0.12299 named in the title, and the outlined best cell loses to it**, paired $\Delta -0.02027$ $[-0.03494, -0.00697]$, CI below zero. The dual arm that merely *ties* the baseline is the fusion_layers = 0 variant of that same view pair (0.11880, -0.00419 straddling zero), and it is *not* in this matrix — which is itself the point of Section 10.3: at fixed views, removing the fusion head is worth more than any choice of views. *Source: 2026-07-25-nexttrack-dual-tower-fusion-scan.md.*

adds zero functions to the hypothesis space. The only thing the pre-MLP actually changes is the ReLU, and the ReLU is applied to PCA coordinates, which are zero-centred and near-symmetric *by construction*: about half of every input vector’s coordinates are negative, and half-wave rectification sets every one of them to exactly zero. Sign is not recoverable afterwards, so this is a permanent deletion, not a reparametrization. A ReLU layer is lossless on signed input only if it is over-complete enough to re-encode the negative half-space — canonically $\text{concat}[\text{relu}(x), \text{relu}(-x)]$ at $2\times$ width. At $\text{pre_hidden} = 256$ on a 192-dimensional input the ratio is only $1.33\times$, so after rectification the layer carries *fewer live coordinates than the input has dimensions*. Figure 21 draws exactly this.

Results. The decisive single-tower ablation loses: MLP256→GRU256 scores 0.10901 (156 of 1431) against the baseline’s 0.12299 (176), paired $\Delta -0.01398$ $[-0.02657, -0.00140]$, CI below zero (Figure 23). The mechanism was then *measured* by a counterfactual tap — the model’s own affine map with every ReLU deleted. Next-item genre decodability (11 184 probe test rows, 25 classes) reads raw latent 0.4033 accuracy / 0.7609 AUC → pre-linear 0.4084 / 0.7667 (the affine map is *neutral*, $\sim 0.8\sigma$, exactly as $W_i(Vx) = (W_iV)x$ predicts) → pre-ReLU 0.3700 / 0.7508 (the ReLU is the *whole* loss, $\sim 5.9\sigma$). The zeroing is exact and quantified: **frac_exact_zero** on the rectified tap is 0.5327, matching **frac_negative_coords** on the un-rectified tap to four

digits, and $256 \times (1 - 0.5327) = 119.6$ live units to carry 192 signed directions. The damage survives the recurrence: within one model, the tower fed rectified input reads 0.3929/0.7845 against its identical twin fed raw input at 0.4183/0.8004 ($\sim 3.9\sigma$), and the raw tower reproduces the standalone plain GRU (0.4235/0.8006). The repair signature confirms the direction of causation: adding a *bare* tower alongside the pre-encoded one recovers +0.01468 [+0.00278, +0.02655] over the single pre-encoded model yet only *ties* the plain baseline (+0.00070 [-0.01188, +0.01328]) — it restores an un-rectified path, it never exceeds it. Dose is flat and nominally negative throughout (pre_hidden 0/128/256/384 $\rightarrow$ 0.10273/0.09993/0.09713/0.09993, every adjacent pair straddling zero, no interior peak), and placement is irrelevant — rectifying both towers ties rectifying one.

The honest confound. The pre-MLP ends in Dropout, so pre-encoder arms receive GRU-*input* dropout that the bare arms lack (the baseline drops the recurrent output only). Part of the measured harm may therefore be over-regularization. A `pre_dropout` parameter to disentangle it is identified but not implemented.

Source: 2026-07-26-nexttrack-pre-encoder-scan.md.

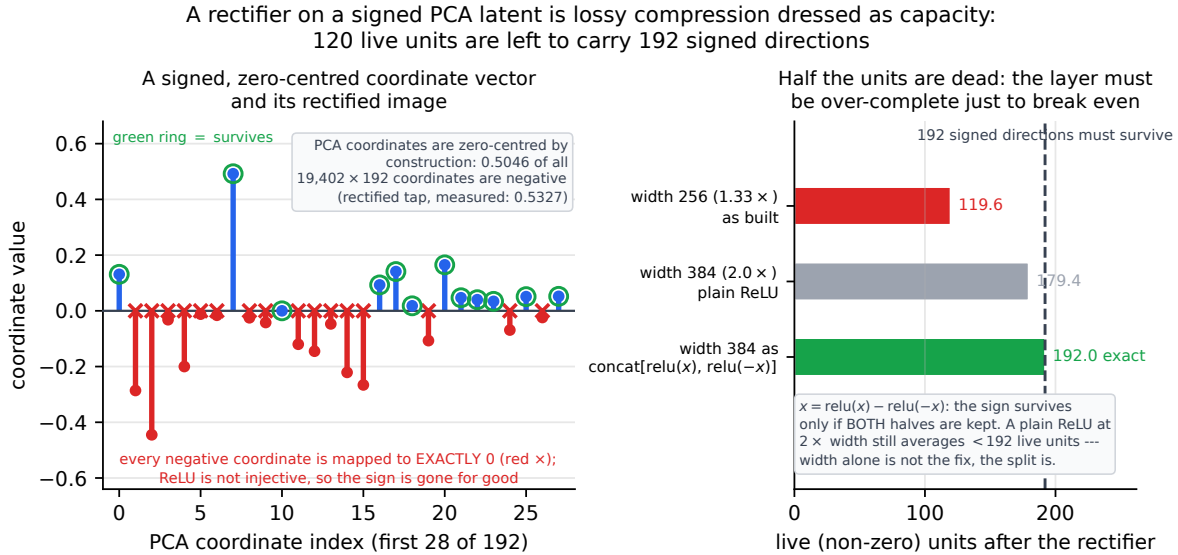


Figure 21: A rectifier on a signed PCA latent is lossy compression dressed as capacity. *Left*: one real item’s first 28 PCA-192 coordinates (blue positive, red negative) with its rectified image overlaid — every negative coordinate is mapped to exactly 0 and, ReLU not being injective, the sign is unrecoverable. 0.5046 of all $19\,402 \times 192$ coordinates in this item space are negative by construction, and the measured rectified *tap* inside the trained model reads 0.5327 (the two are kept distinct: the tap sits after a learned affine map). *Right*: the width budget — at the built width 256 only 119.6 units stay live to carry 192 signed directions, a plain ReLU at $2 \times$ width still averages under 192, and only the concat[relu(x), relu($-x$)] split is exact, since $x = \text{relu}(x) - \text{relu}(-x)$ needs both halves. This panel is a budget, not a run measurement. It complements Figure 22, which measures the same mechanism as decodability rather than drawing it.

10.3 The fusion head is actively harmful, and the harm grows with each upstream rectifier

Rationale. Why would extra capacity between the encoder and the readout *cost* accuracy? Because the retrieval geometry is fixed and external to the network. Training is in-batch InfoNCE over

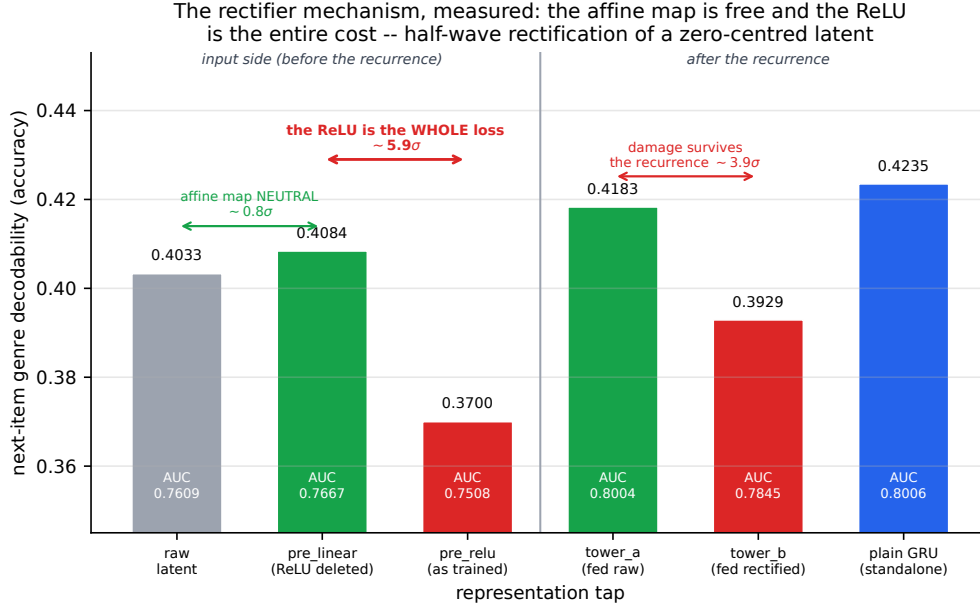


Figure 22: The rectifier, not the extra layer, is the whole loss. Next-item genre decodability (probe accuracy; AUC printed inside each bar) across six representation taps, 11 184 probe test rows over 25 classes, paired SE ≤ 0.0065 . Input side, left of the divider: raw latent (gray) $\rightarrow$ the same map with every ReLU *deleted* (green, 0.4084, neutral at $\sim 0.8\sigma$) $\rightarrow$ the map as trained (red, 0.3700, $\sim 5.9\sigma$ worse). After the recurrence, right of the divider: the tower fed raw (green, 0.4183) against its identical twin fed rectified (red, 0.3929, $\sim 3.9\sigma$), with the standalone plain GRU (blue, 0.4235) reproducing the raw tower. The affine map is free; the half-wave rectifier is not, and its damage is still present after the GRU. **Two reading cautions.** The y axis is truncated to roughly 0.355 to 0.445 with no break drawn, so the bar *heights* exaggerate what are 3 to 8% relative differences in probe accuracy; the claim rests on the annotated σ multiples, not on the visual drop. And a probe accuracy of 0.4033 on 25 classes is far above chance but far from perfect — these are decodability *differences*, not a statement that genre is fully recoverable from a latent. *Source: 2026-07-26-nexttrack-pre-encoder-scan.md.*

$L2$ -normalized vectors and serving ranks the catalog by cosine to the predicted 192-dimensional vector, so the readout's only job is to land in a *direction* in a signed linear space it does not control. With `fusion_layers = 0` the map from $\mathbf{R}^{512}$ to $\mathbf{R}^{192}$ is a single unconstrained linear operator — already the most general linear read of the recurrent state and exactly matched to the geometry it must hit. Inserting `Linear(512 $\rightarrow$ 256)  $\rightarrow$  ReLU  $\rightarrow$  Dropout  $\rightarrow$  Linear(256 $\rightarrow$ 192)` does not restrict which directions are *reachable*: the final map is unconstrained and $256 > 192$, so the image of the non-negative orthant of $\mathbf{R}^{256}$ under a generic linear map generically *positively spans* all of $\mathbf{R}^{192}$, and no prediction direction is lost. The cost is parametrization and noise, not reachability, and it comes in two parts. First, every direction with a negative coefficient must now be produced by *cancellation* among non-negative activations rather than directly, so the loss spends gradient undoing a warp it never wanted — and the warp is the same half-wave sign deletion as Section 10.2, now applied to the recurrent state instead of to the input. Second, and more concretely, the trailing Dropout injects multiplicative noise directly into the very vector whose *direction* is being scored at retrieval time; a dropped unit does not attenuate the score, it rotates the prediction. Neither part is a claim about expressivity, and the argument's weight rests on the measurement below, which

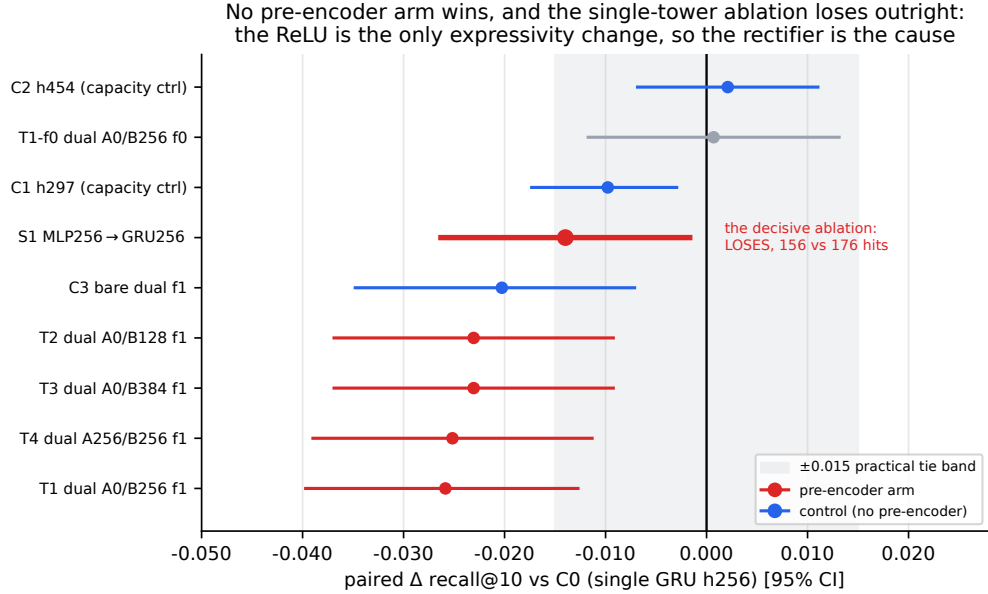


Figure 23: The pre-encoder axis at a glance, and the template for reading every paired comparison in this document. Paired per-session bootstrap Δ Recall@10 against the reproduced single-GRU baseline (2000 resamples, rng 1337, $n = 1431$), nine arms sorted by Δ , with the ± 0.015 practical tie band shaded and a black line at zero. Controls are blue, pre-encoder arms red; the thick row is the decisive ablation, MLP256→GRU256 at -0.01398 $[-0.02657, -0.00140]$ (156 hits against 176). Note what the figure does *not* show: no arm’s CI clears zero on the positive side, which is the shape of every architecture scan on this corpus (Section 11). *Source: 2026-07-26-nexttrack-pre-encoder-scan.md.*

is CI-backed three times.

Results. Three independent paired measurements, all CI above zero in favour of *removing* the head: $+0.01607$ $[+0.00210, +0.03005]$ in the redundant-view regime, $+0.02027$ $[+0.00629, +0.03354]$ in the complementary-view regime, and $+0.02655$ $[+0.01398, +0.03913]$ with a pre-MLP on one tower. The head also *costs* parameters to do this: 871 872 with it against 789 696 without, and 970 432 against 888 256 in the pre-encoder batch — roughly 10 % more parameters for a reliably worse result. *On the ordering, be careful.* It is tempting to read $+0.016 \rightarrow +0.020 \rightarrow +0.027$ as a penalty growing with the number of upstream rectifiers, but the first two arms have *no* upstream rectifier at all — both are bare-tower regimes differing only in which views the towers read — so that step cannot be attributed to rectifier count. All three intervals also overlap heavily and no pairwise test between them was run. What is defensible is weaker and still useful: the penalty **reproduces three times across three different regimes**, and the largest of the three is the one arm that does carry an upstream rectifier. The ordering itself is nominal.

Interpretation. The unifying observation across the whole family is checkable in the sources: *the winning path contains zero half-wave rectifiers*. A GRU (sigmoid/tanh gates, sign-preserving and bounded) plus one `nn.Linear` to the latent dimension; and the single largest gain on record, the champion’s content projection, is a bias-free *linear* map with identity initialization trained by InfoNCE. Every arm that inserts a ReLU into that path loses — the pooled ANN twin, which is a three-layer ReLU MLP, loses to the same-space GRU by -0.0245 $[-0.039, -0.011]$. One counter-

direction must be stated honestly: under the *old* three-factor crown the `fusion_layers = 1` arms scored *higher*, because the extra warp diffuses the top ten (raising `ild`, lowering `artist_adj`) at the cost of exact hits. That apparent advantage disappears under the grounded composite, where no pre-encoder arm reaches ΔH CI above zero at all.

Sources: `2026-07-25-nexttrack-dual-tower-fusion-scan.md`, and `2026-07-26-nexttrack-pre-encoder-scan.md`.

10.4 Four learned combiners, four nulls, and an oracle that says there is nothing to learn

Rationale. The champion’s fusion weights are hard-coded equal thirds, which looks like an obvious place to improve. It has now been attacked from four directions and all four fail, and the reason is best stated in the order that makes the weaker argument unnecessary. **First: there is nothing to learn.** Fit the three blend weights directly on the *test* labels — outright cheating, and therefore an upper bound on what *any* weighting procedure could achieve — and the result is statistically indistinguishable from equal thirds. If the optimum is a tie with the constant, then every parameter a learned combiner spends is pure variance with no bias left to remove, and no amount of data or cleverness changes that. **Second, and only then: even if there were something to learn, it could not be fitted here.** The count-based legs are built *from* the train sessions, so on any train-derived validation slice they look near-perfect — they are being scored on data they memorized — and a weight-fitter therefore learns “trust the count leg”, which is exactly wrong on a test distribution where 55 % of targets have no counts at all.

Results. (1) The XGBoost stacker scores below its *own* GRU base leg in every arm, on both splits, and each Δ is against the base leg on the *same* split: on the canonical PCA-192 split 0.08526 against 0.12299, paired Δ -0.0377 $[-0.056, -0.020]$; on the PCA-128 split `seq-20260715-030509` 0.0867 and 0.0797 against that split’s GRU anchor of 0.1146, paired Δ -0.0280 $[-0.045, -0.011]$ and -0.0349 $[-0.053, -0.017]$. All three CIs are below zero. A combiner scoring *below its own input* needs an explanation, and there are three separable ones. Its training matrix emits exactly *one* positive per train session plus 50 popularity-sampled negatives: 5723 positives against the GRU’s 73 632 teacher-forced steps over the same sessions, $12.87\times$ less signal. Its candidate distribution at fit time is one true item against 51 popularity-sampled candidates — 0.258 % of the catalog — and at eval time it must rank all 19 402. And a 200-round depth-6 booster is piecewise *constant*, so ranking 19 402 candidates with a step function produces large tied blocks broken arbitrarily by item index, which destroys ordering information the GRU leg already had; the z-blend by contrast is a strictly monotone continuous function of the legs’ z-scores and preserves each leg’s internal ordering exactly. (2) Reciprocal-rank fusion, which keeps only ranks and discards score magnitude, scores 0.1572, -0.029 CI below zero — direct evidence that the z-blend’s retention of calibrated magnitude is the working part. (3) The **oracle**: coordinate-ascent weights fitted directly on the *test* labels, an upper bound on any achievable weighting, reach 0.1901, paired Δ $+0.0042$ $[-0.004, +0.013]$, straddling zero. Fitting the same weights on a train-val slice instead collapses to the degenerate vector $[0, 1, 0]$, pure Markov. (4) Representation-level *joint end-to-end* fusion, the strongest form of the standing rebuttal, is Section 10.1: eight arms, a 0.098 % parameter-matched control, best arm a tie, six of eight losing outright.

Interpretation. Put the two halves together and the lane is closed from both ends. The combination weights would have to be estimated from 5723 sessions with one usable event each, while the things being combined were estimated from 73 632 events — and the oracle says the target of that estimation is statistically indistinguishable from the constant. So the z-blend wins because it is a *zero-parameter* estimator of a quantity whose true value is (indistinguishably) equal thirds.

The adopted corollary is that no further learned combiner should be proposed without a mechanism that is not capacity, not co-adaptation and not score-level fusion. The residual uncertainty is scope, not validity — these nulls are established on one corpus with one very unusual property (a single user, 55 % cold) and should not be generalized to session recommendation at large.

Sources: `2026-07-14-nexttrack-blend-and-ensemble.md`, `2026-07-15c-nexttrack-phase2-model-sweep.md`, `2026-07-18-nexttrack-literature-fit-campaign.md`, and `2026-07-25-nexttrack-dual-tower-fusion-scan.md`.

10.5 Depth, width, direction: a spent lever, a weak instrument, and an objective artifact

Rationale. Sessions here average 13.84 events, so the recurrence’s job is to integrate ~ 14 steps of a signed 192-dimensional latent into one direction — a short-horizon integration problem, not one that wants a deep feature hierarchy. Stacking layers buys hierarchy the task does not need while inserting more saturating stages between input and readout and adding inter-layer dropout.

Results. Depth is flat then falls off a cliff: 1-layer 0.0894, depth-2 0.0922 (flat, in-band), depth-3 0.0825 — the only depth arm to lose to the first-order Markov bar, paired Δ -0.0245 $[-0.043, -0.006]$. A residual connection was never *isolated*: the only residual arm is depth-2 at $h = 256$ (0.0908) and no depth-2 $h = 256$ arm was run *without* it, so the honest statement is only that the residual arm fails to beat the depth-2 $h = 128$ arm (0.0922) at twice the width — not that residual connections were tested and found neutral. Width is likewise closed on the rich space — $h = 425$ at $2.2\times$ the parameters ties $h = 256$ — *but the instrument that established it is weaker than it looks*, and that caveat matters more than the result. The four PCA-192 width points read 176/162/ 173/179 hits for $h = 256/297/425/454$: non-monotone, spanning 0.0119 recall with no trend, and the $h = 297$ control actually *lost* to $h = 256$ (-0.00978 $[-0.01747, -0.00280]$, CI below zero) while $h = 454$ tied it. So between-configuration jitter is $\sim \pm 0.01$ — roughly two-thirds of the tie band — while *same*-configuration reproduction is bit-identical. “Width is flat” survives as a trend claim resting on the endpoints tying, not as a per-width measurement, and a single unreplicated param-matched control cannot carry a verdict. The prescribed replication (two runs on the second leak-free split; the path is deterministic, so this needs a second *split*, not a second seed) has not been bought.

The bidirectional arm, and why it is not a directionality result. The record’s cleanest example of a confound that looks like a finding. A backward pass would let step t read item $t + 1$ and copy its own target, so the bidirectional arm could not use per-step teacher forcing at all and had to switch to a leak-free *last-item* objective: encode the whole prefix, predict only the held-out next item. That reduces supervision from ~ 13 targets per session to 1 — the same 73 632 against 5723 arithmetic that is the first mechanism behind the stacker, which is why the two worst designs on record share a cause. The measured collapse (Recall@10 0.0482 $[0.037, 0.059]$, MRR 0.0225, paired Δ versus the Markov bar -0.0587 $[-0.077, -0.041]$, the largest single-arm harm on record) is therefore attributable to the *objective*, not to directionality, and the report declines to close bidirectionality: a fair test would need a different leak-free scheme such as masked-item training. The two-tower family excludes bidirectional mode by construction, with the leakage argument written into its module docstring.

Sources: `2026-07-13-nexttrack-topology-scan.md`, `2026-07-25-nexttrack-dual-tower-fusion-scan.md`, and `2026-07-26-nexttrack-pre-encoder-scan.md`.

11 Power: at $n = 1431$ the nulls are upper bounds, not proofs

Rationale. Every null in Section 10 must be read against the resolution of the instrument that produced it, or it becomes a false closure.

Results. At this split’s 1431 test sessions the paired-bootstrap CI half-width is ~ 0.0124 on Recall@10 (the mean of the nine paired comparisons in the pre-encoder batch; 0.0128 for the nine in the dual-tower batch) and ~ 0.0018 on holisticness@10. Against that, an entire architecture family spans very little: the ten-arm pre-encoder batch spans 0.02796 in recall (0.09713 to 0.12509) and 0.003147 in grounded holisticness. Spread over resolution is therefore $0.02796/0.0124 \approx 2.3$ on recall and $0.003147/0.0018 \approx 1.75$ on the crown. Read that ratio as a count of *separable tiers*: if the whole family spans 2.3 half-widths, then at best about two groups of arms can be told apart, and about one on the crown — a heuristic, not a theorem, but one the direct count bears out and then some. Those ten arms give nine adjacent gaps, and *zero* of the nine adjacent recall gaps and *zero* of the nine adjacent crown gaps exceed their own half-width: only the batch *endpoints* separate at all. In the wider dual-tower batch only two of nine adjacent recall gaps clear it. Resolving a 0.005 recall difference — a size chosen here as illustrative of a difference one would want to act on, not as a pre-registered threshold — would need $1431 \times (0.0124/0.005)^2 = 8801$ test sessions, $6.2\times$ the current split, or roughly 44 000 sessions of listening history at the split’s 20 % test fraction. Figure 24 draws the curve.

Interpretation — and exactly which claims the caveat applies to. The defensible statement is that **large harms were detected reliably** — the pre-encoder’s -0.0140 , the fusion-head penalty’s $+0.0266$, depth-3’s -0.0245 , the bidirectional arm’s -0.0587 , the stacker’s -0.038 to -0.028 , the degenerate centroid arm’s -0.0566 , all with CIs excluding zero — while **moderate wins were never detectable at all**. Those two halves must be kept apart, because the phrase “the nulls are power-limited” is easy to over-apply. Three of the five refuted axes produced *detections*, not nulls: the pre-encoder’s harm, the fusion head’s harm and depth-3’s harm all have intervals excluding zero, and nothing about the instrument’s resolution weakens them. What is power-limited is a different and narrower claim: the *absence of a win* on each axis. Every one of the five axes ends in “no arm cleared zero on the positive side”, and that sentence is an upper bound on the available gain, not a demonstration of zero — the pre-encoder axis, for instance, is bounded at a tie ($+0.0007$ for its best arm) and no more. The same distinction applies to the *tie* verdicts, which are the genuinely underpowered ones: the dual-tower family’s best arm and the width closure both rest on straddling intervals that a larger split could resolve either way. It is *not* “these architectures do not work”.

The nuance that must travel with the caveat. The ~ 0.0124 figure is a *cross-model* floor: it applies when two arms have separately trained weights, because then almost every test session can change its outcome. When two arms share *bit-identical* trained weights and differ only in an eval-time re-rank, the pairing is far tighter and the instrument is 3 to $33\times$ sharper — across the eight MMR arms the *measured* half-widths are $\text{hw}(\Delta H)$ 0.000382 to 0.002476 and $\text{hw}(\Delta \text{recall})$ 0.004193 to 0.018169. The reason is visible in n_{disc} , the number of sessions whose hit@10 changed at all: as low as 9 at $\lambda = 0.9$ against 188 at $\lambda = 0.3$, because re-ranking the top of a candidate pool only flips sessions whose true item crosses the $k = 10$ boundary. That sharpness overturned a pre-registered call — one arm budgeted as “likely unresolvable” at 0.6 to 0.8 conservative half-widths in fact resolved at 4.8 *measured* ones. Two riders: measured half-widths *grow* with distance from the control (0.000382 at $\lambda = 0.9$ to 0.002476 at $\lambda = 0.3$), so the instrument is sharpest exactly where the effects are smallest; and holisticness@10’s absolute scale differs $\sim 3\times$ between models (0.0018 is 21 % of the champion blend’s H but only 7 % of the single GRU’s), so the crown

instrument is intrinsically $\sim 3\times$ more precise on the GRU than on the champion. The practical rule this record now follows is to budget the minimum detectable effect from the *pairing structure* of the comparison and to report the measured per-arm half-width rather than a family constant.

Sources: *2026-07-25-nextrack-dual-tower-fusion-scan.md*, *2026-07-26-nextrack-pre-encoder-scan.md*, and *2026-07-26-nextrack-mmr-lambda-frontier.md*.

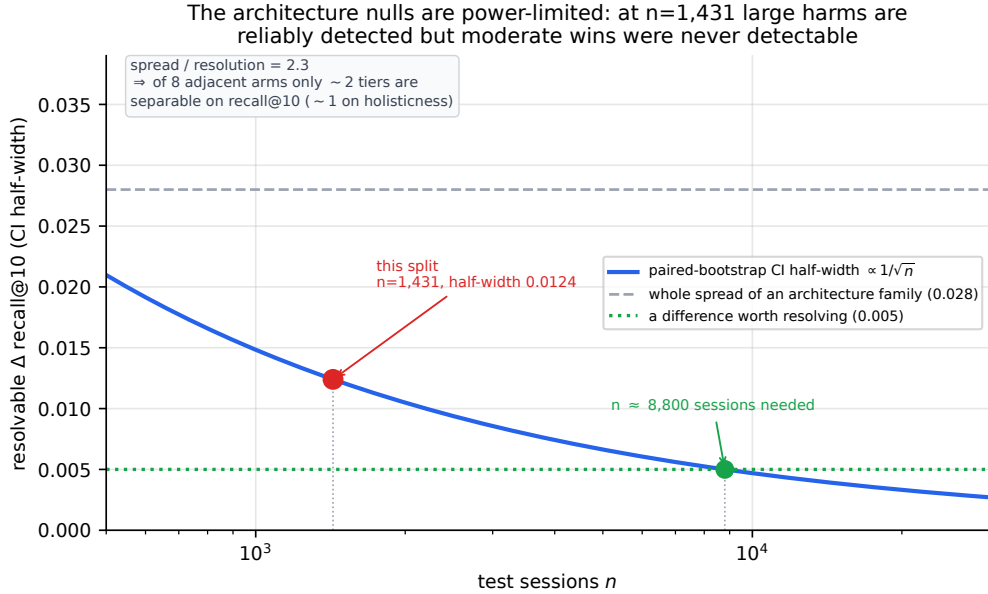


Figure 24: The architecture nulls are power-limited, and this is the figure that reframes Section 10. Paired-bootstrap CI half-width on Recall@10 against test-set size n on a log axis, falling as $1/\sqrt{n}$ and anchored at the measured point $n = 1431$, half-width 0.0124 (red dot). The dashed gray line at 0.028 is the *entire* spread of an architecture family; the dotted green line at 0.005 is a difference one would want to be able to act on — an illustrative choice, not a pre-registered threshold — and it meets the curve only at $n \approx 8800$ sessions, $6.2\times$ this split. The boxed callout gives spread over resolution as 2.3 on recall, from which the “about two separable tiers” follows as a heuristic (tiers $\approx \lfloor \text{spread}/\text{half-width} \rfloor$); the callout’s “eight adjacent arms” is loose — the batch has ten arms and therefore nine adjacent gaps, none of which clears its own half-width. Read with Section 11: this floor applies to *separately trained* arms; shared-weight eval-time comparisons such as Table 3’s are 3 to $33\times$ sharper. Sources: *2026-07-25-nextrack-dual-tower-fusion-scan.md*, and *2026-07-26-nextrack-pre-encoder-scan.md*.

12 Where the gains actually came from

Read as one arc, the record says something narrower and more useful than “the blend is best”. Four claims survive.

Gains came from adding a signal of a different kind, never from more model. The board moved three times: when a co-occurrence lookup was fused with a sequence model (paired $\Delta +0.0657$ over the bar, on the earlier seq-20260713-014226 artifact where that step was taken), when a content retrieval leg was added to that (+0.0147, CI above zero), and when the metric those two content-shaped legs retrieve in was *learned* (+0.0259, CI above zero, hardened to +0.0426 on a colder split). It never moved for width, depth, direction, an extra tower, an extra nonlinearity, or a learned combiner. The mechanism is the cold half: a new *kind* of signal can reach sessions the existing ones cannot, whereas more capacity only re-describes what is already reachable.

The winning path is linear where it touches the retrieval geometry. The loss and the serving metric both live in a fixed external space — cosine retrieval over signed 192-dimensional latents — so the readout should be the most general map *into* that space and nothing more. Every measured insertion of a half-wave rectifier into that path lost: on the input side (−0.0140, with the ReLU isolated as the whole cause at $\sim 5.9\sigma$ of decodability), on the output side (three confirmations across three regimes, at 10 % more parameters), and in the pooled ANN twin (−0.0245, same space, PCA-128). The single largest win, meanwhile, is a bias-free linear map. “Add a head” is a reflex worth resisting whenever the target space is fixed.

The combination should not be learned at this data volume, and that is measurable in one run. An oracle weighting fitted on the test labels buys +0.0042 with a CI straddling zero. That single probe bounds the entire learned-combiner lane, and it should be the *first* experiment run in such a lane, not the last. The transferable discipline is to count the supervision events available to the combiner separately from those available to the parts, and to check whether the candidate distribution at fit time matches the one at inference.

The metrics are part of the result. A composite built from list properties rather than correctness is maximized by a model that has stopped trying, and the ± 0.015 recall floor was the only thing standing between this record and a crown for a model 46 % worse at its actual job. A grounding factor helps and is not sufficient. The counterexample that makes the metric salvageable rather than useless is MMR, which raises three factors *including* the grounding one, at a recall tie — same metric, opposite mechanism.

13 What is not established

This section exists so that no reader has to reconstruct the caveats from the prose.

Nothing from 2026-07-26 is promoted, and the champion is unchanged. Six MMR definitions were *registered*; the auto-generated `best-<predictor>-<run>` records that exist for those runs are best-models group membership, not deliberate promotion. Of 77 model records only *five* are deliberate promotions, and the recall champion remains $R'+M+C'$ at 0.21174. The best-models group currently holds nine of twelve slots, all nine marked `source: auto`. Eight of the nine come from the 2026-07-26 batches and the ninth (the two-leg R+M blend) from 2026-07-22, and two of them — the TIER-2 *priced* arms B3 and A2 — have a Δ recall CI below zero. The group is nonetheless *curated*, not an uncurated flood: `GET /api/best-models` returns per-entry `rank`, `metric`, `metric_value`, `source` and `selected_at`, plus a top-level list of 3 `pinned` model names and 75 `excluded` identifiers — and the two TIER-3 *rejected* MMR arms are explicitly among the exclusions. What remains unverified is only whether the current membership reflects a deliberate curation pass or the server’s automatic recompute; `PROJECT-FACTS.md` records that remote-worker completions do not fire that recompute, so the group can be silently stale after any offloaded batch and its `updated_at` must be checked against the newest run’s `finished_at` before it is trusted.

Several nulls are underpowered rather than refuted. See Section 11. In particular the width closure rests on endpoints tying with a jitter of $\sim \pm 0.01$ between neighbouring widths, one control losing outright, and a prescribed two-run replication that has not been bought; and the pre-encoder result carries an unresolved dropout confound.

Several comparisons are nominal, single-run, or report-provenance only. No paired CI exists for the temperature cells, the projection-rank ablation, the BPR cells, the anti-eager β arms, or the canonical-split LSTM — the reports call the rank-64 and LSTM gaps ties and the BPR gap a collapse, but none is backed by a paired bootstrap. The champion’s R' and C' standalone figures (0.1726, 0.1516) exist only as off-server-driver numbers in a campaign report, with no server run and no raw hit counts; only the Markov leg has a real canonical-split run. The warm/cold decompositions, the hit-set overlap counts, and the candidate-geometry measurement of Figure 19 are single-run observations with no uncertainty quantified. Two β arms executed on a stale worker checkout and are not code-comparable to any current batch.

The crown metric is young and has already been revised twice, and its grounding floor 0.34 is currently operating *outside* the regime it was calibrated for: it was set to cancel gains bought by destroying recall, and MMR is the first lever that moves music@10 in the opposite direction. Stored holisticness values are comparable only within a generation (Table 1), the generation labels used here were re-derived from stored per-row facets rather than read from the API, and two runs store gen-3 values whose re-derivation is off by $\sim 1 \times 10^{-3}$ because their `artist_conc_at_k` column is missing from the metrics index — their H is trustworthy, their facet row is not, and their complete twins were quoted instead.

seq-embed has no numbers at all, `seq-ann` has no canonical-split run, and no parameter count is recorded anywhere for `seq-stacker` or `seq-ann` (the stacker’s is not even well defined — 200 boosted trees at depth 6 plus a base GRU). Three parameter counts in Table 2 were derived by instantiating the real classes at `latent_dim = 192` rather than transcribed: the LSTM’s 510 144 and the champion’s 431 808 ($= 394\,944 + 36\,864$, the GRU plus the shared projection W).

14 Conclusions

Six trained predictor families on one leak-free next-track split converge on a clear story. (1) The task’s defining property is that 788 of 1431 test targets are *cold*, capping every seen-items-only mechanism at Recall@10 0.449336 and making retrieval in a shared content geometry the only route to the majority of the test set. (2) The champion is a fixed score-level blend, $R'+M+C'$ at 0.21174 (303 of 1431), and it wins because its legs are correct on almost *disjoint* sessions — the Markov leg reads warm 0.2255 / cold 0.0102, the GRU leg is flat across the split, and Jaccard overlap between their hit sets is 0.201. (3) Fusion of unlike signals pays and fusion of like signals does not: a second neural leg contributes nothing, a session-kNN leg is actively harmful (−0.038, CI below zero), and adding a leg only helps when it reaches sessions the others cannot. (4) The single largest win is a learned InfoNCE projection of the item space, +0.0259 [+0.012, +0.041] over the unprojected crown, seed-robust across three seeds and *wider* on a colder second split (+0.0426 [+0.0280, +0.0573]); it lifts both legs that share the geometry (content-kNN 2.2×, with *zero* trainable parameters of its own) and leaves the ID-based leg untouched, and full rank ties rank 64 while BPR collapses 0.055 below InfoNCE — so the gain is the *supervision*, not the capacity. (5) Recurrence beats pooling (−0.0245 [−0.039, −0.011] for the pooled twin, same space, byte-identical objective), and the damage concentrates on the cold half, where the quality of the predicted point is all there is. (6) Eval-time MMR re-ranking is the first honest crown lever — eight of eight arms lift *holisticness@10* with paired CI above zero, *four* of them at a recall tie (three on the plain GRU, one on the champion), and it de-concentrates artists 1.59 to 3.06× faster than it reduces seed echo with no notion of artist in the rule — but it benefits the plain GRU (H 0.0342) far more than the champion (0.0153), whose mood coherence falls twice as fast when diversified at matched λ (−8.5 % against −4.2 %) and collapses −40 % when pushed to $\lambda = 0.3$, and **nothing from that campaign is promoted**. (7) Five design axes are refuted with measured mechanisms: a second tower’s input is measurable with respect to the first’s, so it can add ensemble diversity but no information — and joint training destroys exactly that diversity; a ReLU pre-encoder is provably expressivity-neutral in its linear part and deletes 0.5327 of a zero-centred latent in its nonlinear one, leaving 119.6 live units for 192 signed directions; an MLP fusion head is harmful three times over — reproduced in three different regimes, though the apparent ordering among the three is nominal — at 10 % more parameters; four learned combiners land at or below their own base legs while an oracle weighting fitted on the test labels buys only +0.0042, CI straddling zero; and depth, width and direction are spent, with the bidirectional collapse attributable to the 12.87× supervision cut its objective forced, not to directionality. (8) The record carries two primary metrics and must not silently pick one: Recall@10 is the comparability currency, *holisticness@10* is the server’s declared primary, the two orderings are near-inverted, and the crown has had *three* incompatible generations whose values differ by up to 6.4× on identical predictions. (9) Under the live crown the ten separately-trained architecture arms occupy a band 0.02127 to 0.02487 whose whole width (~ 0.0036) is barely twice $\text{hw}(\Delta H) \approx 0.0018$, so their rank order carries no information — while the recall champion, at 0.008603 far below that band, and the MMR arms, at 0.026761 to 0.034171 above it, are *outside* it and separated for real. (10) What is power-limited on all five refuted axes is the *absence of a win*, not the harms: at $n = 1431$ the paired half-width is ~ 0.0124 on recall against a whole-family spread of 0.028, zero of nine adjacent gaps in the last batch exceed their own half-width, and resolving 0.005 would need 8801 test sessions — so large harms were detected reliably with intervals excluding zero, moderate wins were never detectable, and each axis’s closure bounds an available gain rather than demonstrating zero. The genuinely underpowered verdicts are the *ties*: the dual-tower family’s best arm and the single-GRU width closure. (11) Shared-weight eval-time

comparisons are a different instrument entirely, 3 to $33\times$ sharper because n_{disc} can be as low as 9 of 1431, so the minimum detectable effect must be budgeted from the pairing structure and not from a family constant. (12) What actually produced every gain on this board was a new *kind* of signal or a better shared *geometry*, never more parameters, more towers, more nonlinearity, or a learned combination — which is also why **seq-embed**’s per-item table is the one untried family the data budget already argues against, at a median of one observation per observed item.

15 Source reports

This comparison consolidates fifteen next-track campaign reports from `experiments/`, plus the living facts file:

- `2026-07-13-nexttrack-sequence-sweep.md` — establishes the session GRU with in-batch InfoNCE as the single-model workhorse and the first-order Markov lookup as the reference bar (Section 5).
- `2026-07-13-nexttrack-topology-scan.md` — depth, width and direction closed on the early item space: depth-3 **loses** to the bar (-0.0245 , CI below zero), residual was never isolated at matched width, and the bidirectional arm’s collapse to 0.0482 is an *objective* artifact (Section 10.5).
- `2026-07-14-nexttrack-blend-and-ensemble.md` — the fixed z-blend of a sequence model with a co-occurrence lookup, $+0.0657$ [$+0.049$, $+0.083$] over the bar, three-seed confirmed; a second neural leg is redundant (Section 6).
- `2026-07-14-nexttrack-graded-relevance-and-new-legs.md` — introduces `artist@k / genre@k / artist_mrr` as first-class graded metrics (Section 3).
- `2026-07-15-nexttrack-compression-representation-bakeoff.md` — the compression/representation bake-off.
- `2026-07-15b-nexttrack-compression-dim-extension.md` — with the dimension extension, these two fix PCA-192 as the shared item space every trained model reads (Section 1).
- `2026-07-15c-nexttrack-phase2-model-sweep.md` — the family sweep: the pooled ANN twin loses to the same-space GRU (-0.0245 , CI below zero) and the XGBoost stacker lands below its own base leg (Sections 8 and 10.4).
- `2026-07-18-nexttrack-literature-fit-campaign.md` — the learned InfoNCE projection champion, $+0.0259$ over the prior crown, plus the oracle-weight probe that bounds the learned-combiner lane at $+0.0042$, CI straddling zero (Sections 7 and 10.4).
- `2026-07-18b-nexttrack-projection-registration-and-scan.md` — registers the champion and scans its axes: InfoNCE $\gg$ BPR, full rank $\approx$ rank 64, τ mildly monotone with the default **held** (Section 7).
- `2026-07-18c-nexttrack-representation-exploration.md` — six of eight alternative item spaces lift the *frozen* readout with CI above zero and *none* lifts the champion: whitening and the learned projection are the same fix in two places (Section 7).
- `2026-07-20-musical-distance-metric.md` — `music@10`, the graded grounding metric that later made the crown salvageable (Section 3).
- `2026-07-22-session-holisticness-and-antieager-levers.md` — the holisticness suite, the weak anti-eager training knob, and the offline λ sweep that transferred to the native arms to four decimals (Sections 3 and 9).

- 2026-07-25-nexttrack-dual-tower-fusion-scan.md — dual-tower fusion **REFUTED**, the fourth combiner null: zero of eight arms clear zero against a single-GRU width control param-matched to 0.098 % of the $f1$ arms, the MLP fusion head is harmful twice over, and five ΔH “wins” all fail the recall floor (Sections 10.1, 10.3 and 3).
- 2026-07-26-nexttrack-pre-encoder-scan.md — the per-step MLP pre-encoder **REFUTED and actively harmful** (-0.01398 , CI below zero), with the rectifier isolated as the whole mechanism at $\sim 5.9\sigma$ of genre decodability, and the first batch in the project’s history to produce *zero* crown candidates (Section 10.2).
- 2026-07-26-nexttrack-mmrlambda-frontier.md — the MMR λ frontier: eight of eight arms lift $\text{holisticness}@10$ with paired CI above zero, four free of any recall cost (three on the GRU, one on the champion), six definitions **registered and none promoted** (Section 9).
- PROJECT-FACTS.md — the living facts file: the noise bands, the power anchors, the promotion ledger and the metric-generation history (Sections 11 and 13).